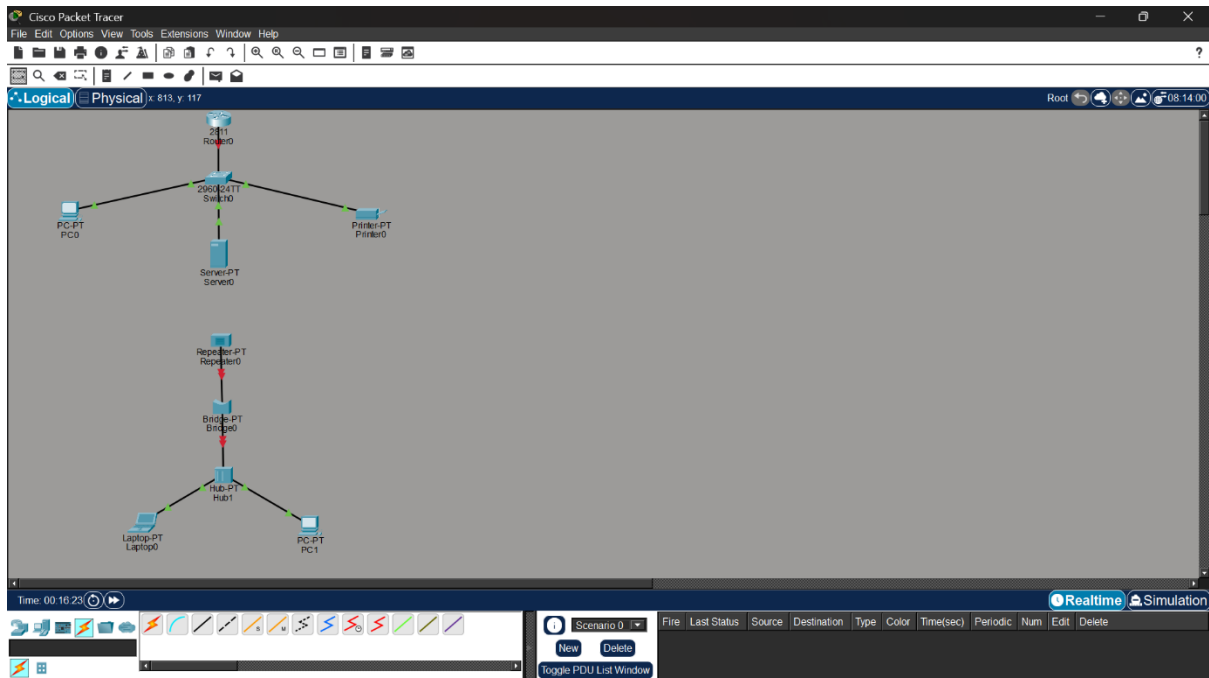
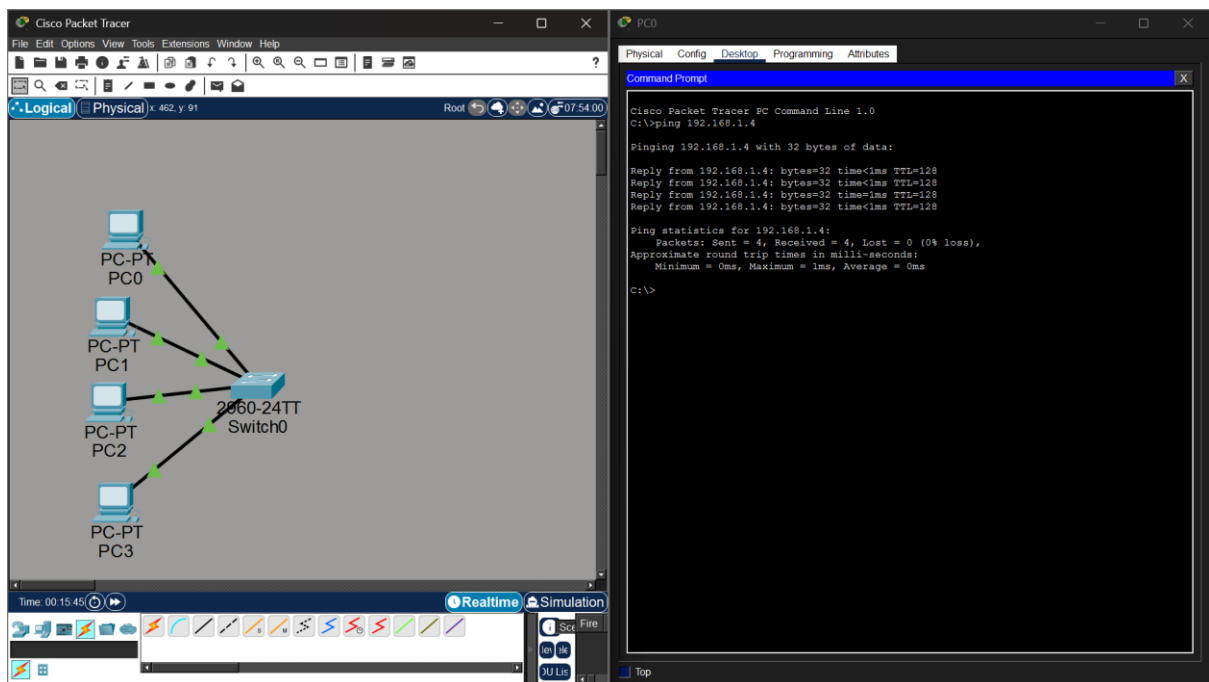


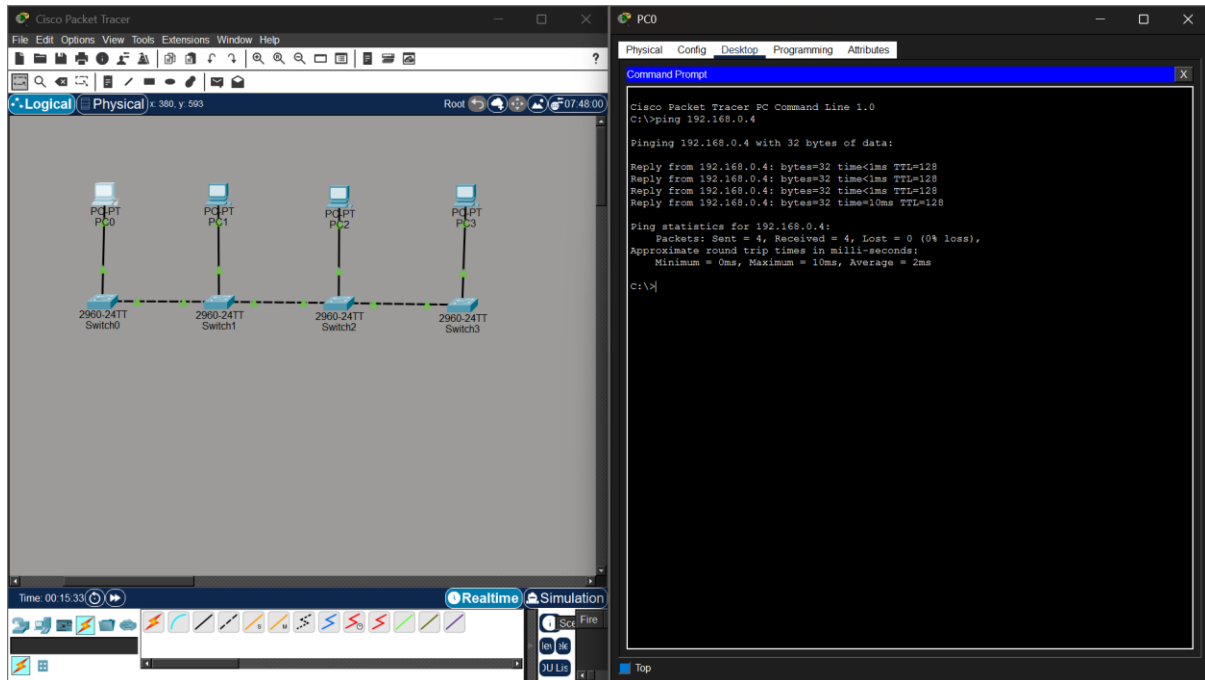
Experiment-01-Network-Components



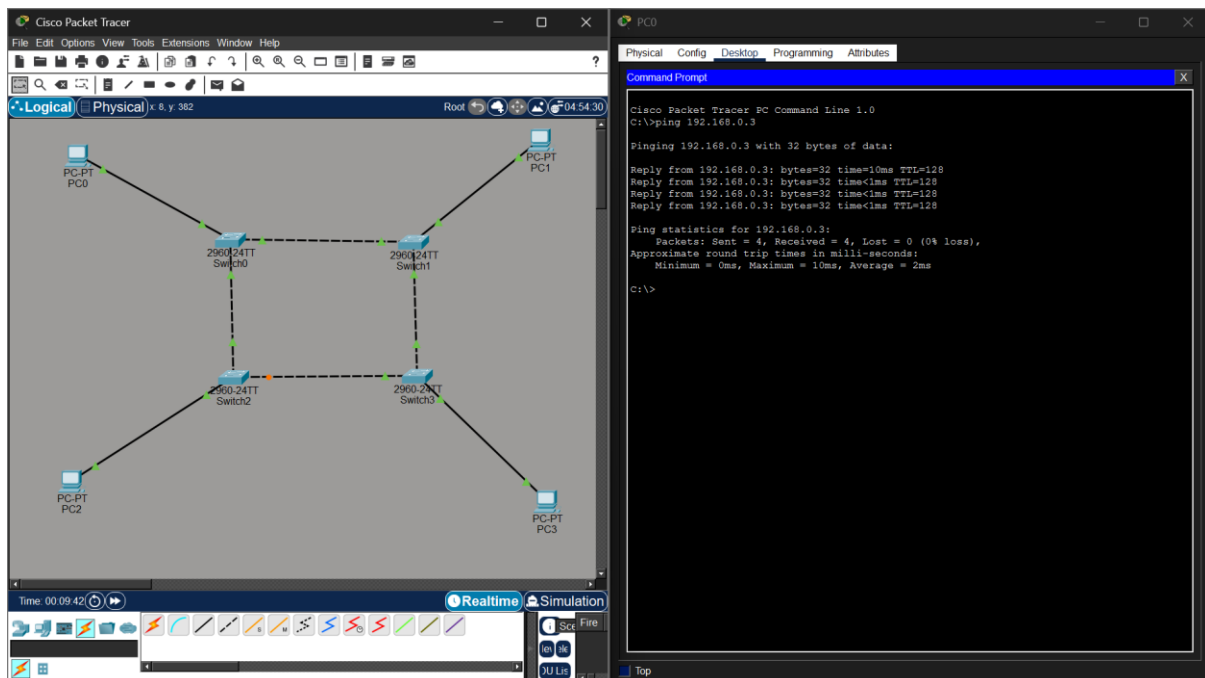
Experiment-02-Star-Topology



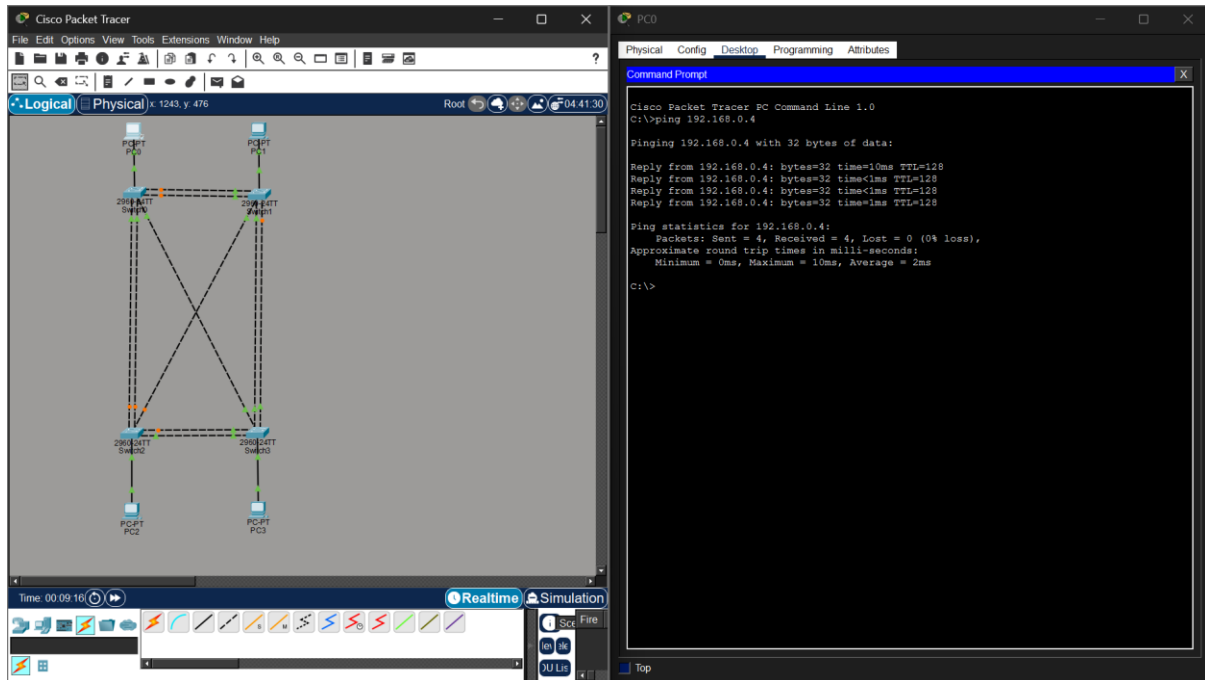
Experiment-03-Bus-Topology



Experiment-04-Ring-Topology



Experiment-05-Mesh-Topology



Experiment-06-Tree-Topology

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1879, y: 983

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
0.005	Switch2	
0.005	PC3	
0.006	Switch2	
0.006	PC4	
0.006	Switch2	
0.007	PC5	
0.007	Switch2	
0.007	Switch0	
0.008	Switch2	
0.008	Switch0	
0.009	Switch0	
0.009	Switch1	
0.010	Switch1	
0.710	--	

Reset Simulation Constant Delay Capturing

Play Controls

Event List Filters - Visible Events

ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT, TCP, LACP, LLDP, MDIOBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIPv2, RTP, SCP, SMTP, SNMP, SSH, SIP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Event List Realtime Simulation

Time: 00:17:34.731 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1539, y: 517

PDU List Window

File	Last Status	Source	Destination	Type	Color	Time/Sec	Periodic	Num	Edit	Delete
Successful	PC0	PC3	ICMP		0.000	N	0	(edit)		(delete)
Successful	PC1	PC4	ICMP		0.000	N	1	(edit)		(delete)
Successful	PC2	PC5	ICMP		0.000	N	2	(edit)		(delete)
Successful	PC6	PC0	ICMP		0.000	N	3	(edit)		(delete)

Time: 00:14:57/

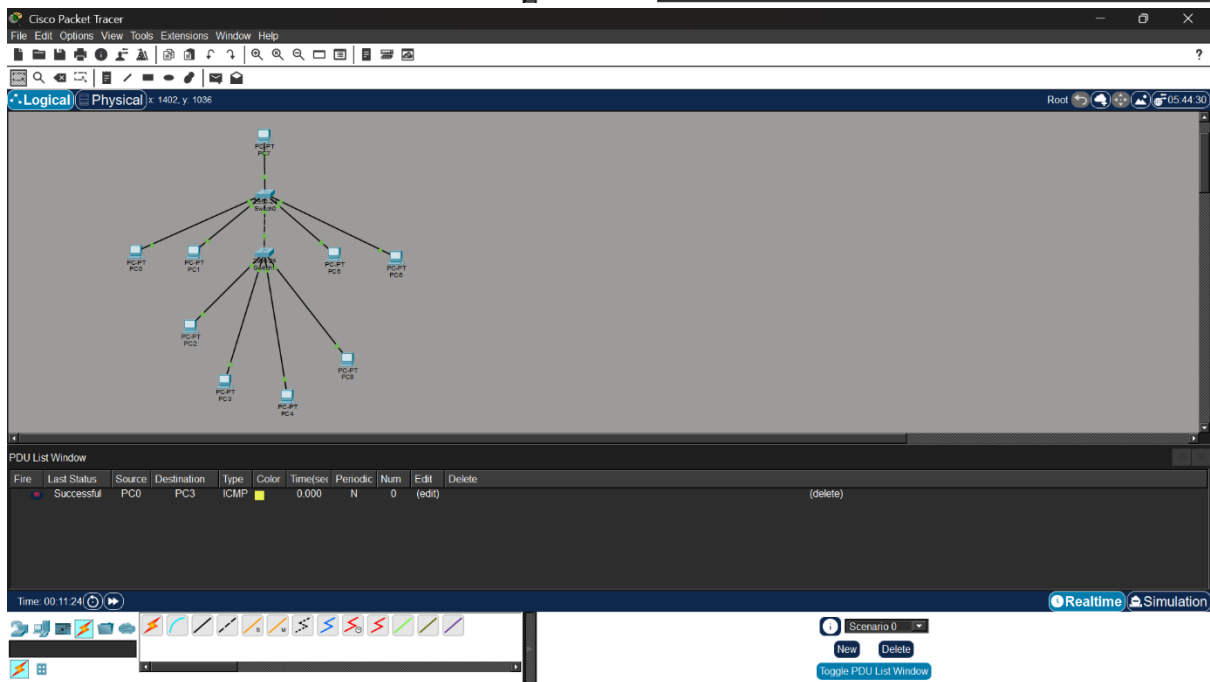
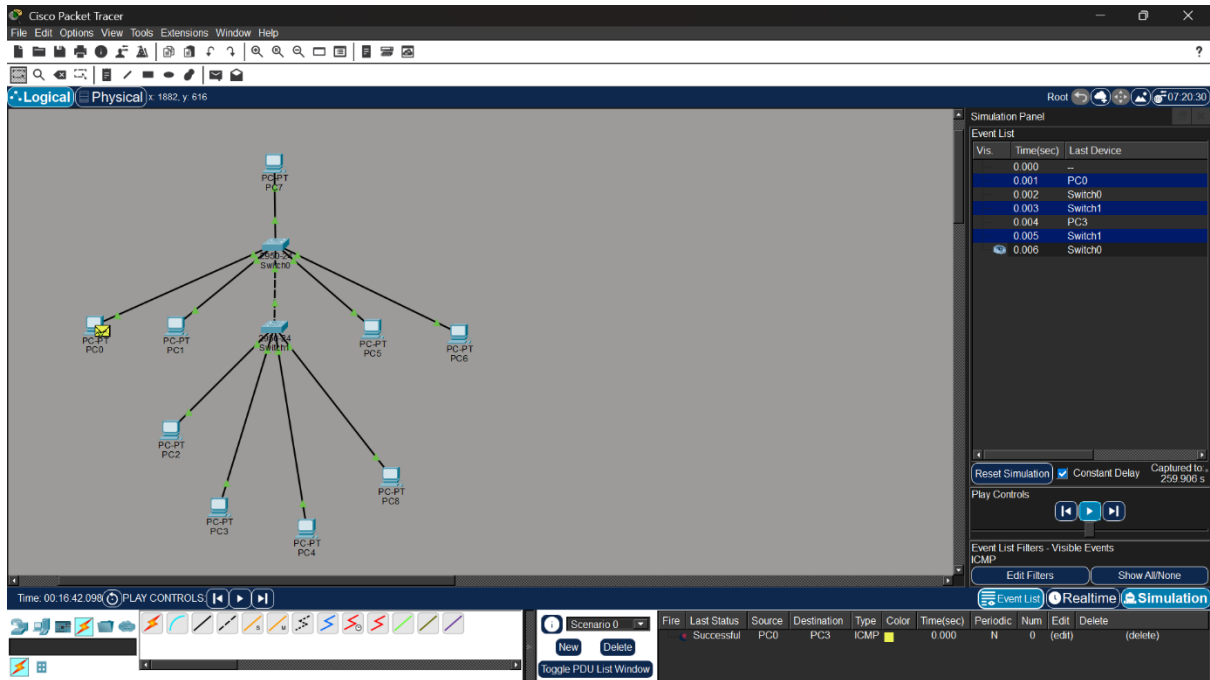
Realtime Simulation

Scenario 0

New Delete

Toggle PDU List Window

Experiment-07-Hybrid-Topology



Experiment-08-ARP-Analysis

The image displays two screenshots of the Cisco Packet Tracer interface, illustrating a network topology and simulation results for ARP analysis.

Top Screenshot:

- Network Topology:** A central Hub0 is connected to six PCs (PC0, PC1, PC2, PC3, PC4, PC5). PC0 is highlighted in blue.
- PDU List Window:** Shows a single entry for a successful ICMP packet from PC0 to PC3 at time 0.000.
- Simulation Controls:** The simulation is running in Realtime mode. The time is 00:12:00.

Bottom Screenshot:

- Network Topology:** The same network topology is shown, but with green checkmarks on the PC icons, indicating successful connectivity.
- Simulation Controls:** The simulation is running in Realtime mode. The time is 01:08:51.264.
- Event List Panel:** Shows a list of events captured during the simulation, including ARP requests and replies between PC0 and Hub0, and Hub0 and PC3.
- PDU List Window:** Shows a single entry for a successful ICMP packet from PC0 to PC3 at time 0.000.

Experiment-09-CSMA-CD-CSMA-CA

The image displays two screenshots of the Cisco Packet Tracer interface, illustrating network configurations and command-line outputs for CSMA-CD and CSMA-CA experiments.

Top Screenshot: The main window shows a network topology with a central switch (Switch0) connected to three PCs (PC0, PC1, PC2). The simulation panel on the right shows the event list with the following entries:

Vis.	Time(sec)	Last Device
0.000	-	-
0.001	PC0	
0.002	Switch0	
0.003	PC1	
0.004	Switch0	

The command prompt window on the right shows the following output:

```
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: FE80::2D0:FFFF:FED5:9268
IPv6 Address...: ::
IPv4 Address...: 192.168.1.1
Subnet Mask...: 255.255.255.0
Default Gateway...: 0.0.0.0

Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: ::
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: 0.0.0.0

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Bottom Screenshot: The main window shows the same network topology. The simulation panel on the right shows the event list with the following entries:

Vis.	Time(sec)	Last Device
0.000	-	-
0.001	PC0	
0.002	Switch0	
0.003	PC1	
0.004	Switch0	

The command prompt window on the right shows the following output:

```
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: FE80::2D0:FFFF:FED5:9268
IPv6 Address...: ::
IPv4 Address...: 192.168.1.1
Subnet Mask...: 255.255.255.0
Default Gateway...: 0.0.0.0

Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: ::
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: 0.0.0.0

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Experiment-10-Computer-Lab

The image displays two screenshots of the Cisco Packet Tracer application. The top screenshot shows a network topology in the 'Logical' view. A central switch, labeled '2960-24 Switch0', is connected to four PC-PT devices (PC0, PC1, PC2, PC3) and a router, labeled '1841 Router0'. The bottom screenshot shows the same network topology, but with a 'Command Prompt' window open on PC0. The command prompt displays the output of the 'ipconfig' command, showing the IPv4 address 192.168.1.1, subnet mask 255.255.255.0, and default gateway 0.0.0.0. It also shows the output of the 'ping 192.168.1.254' command, indicating a successful ping with 0% loss and a round trip time of 12ms. The command prompt window is titled 'PC0' and has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'.

Network Topology:

- Router: 1841 Router0
- Switch: 2960-24 Switch0
- PCs: PC-PT PC0, PC-PT PC1, PC-PT PC2, PC-PT PC3

Command Prompt Output (PC0):

```
C:\>ipconfig

FastEthernet0 Connection (default port)

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::230:F2FF:FETA:2C39
    IPv6 Address . . . . .: 
    IPv4 Address. . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: 
    IPv6 Address . . . . .: 
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time=12ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255

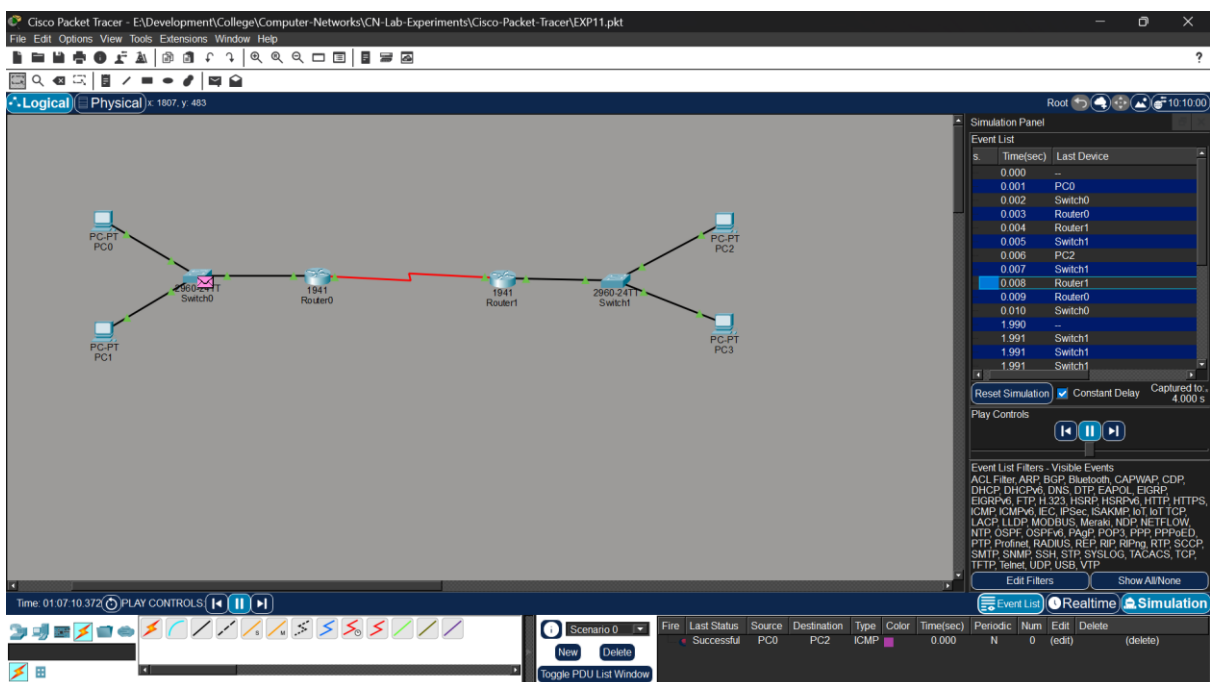
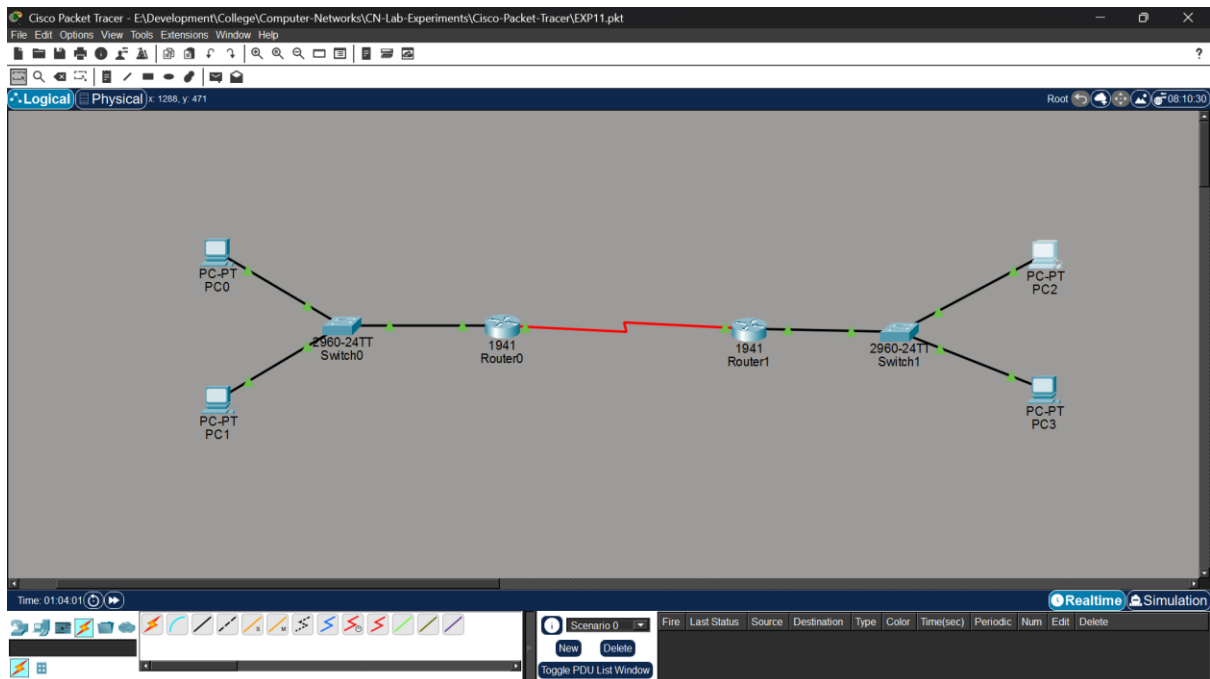
Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

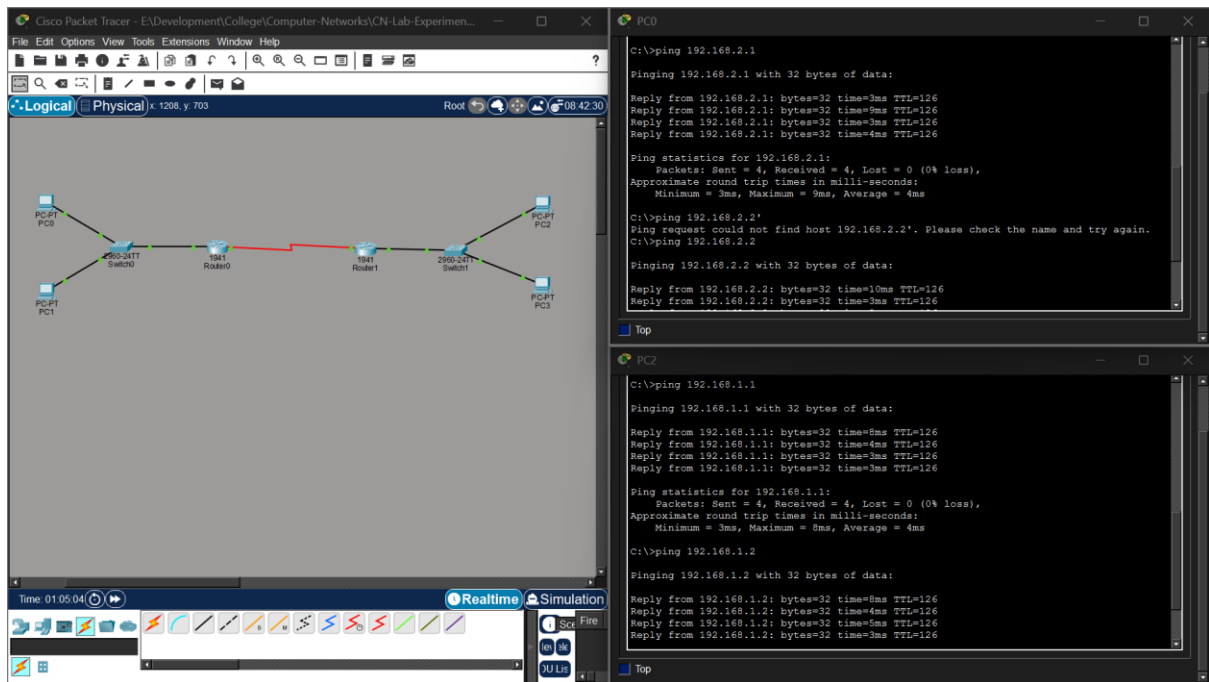
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

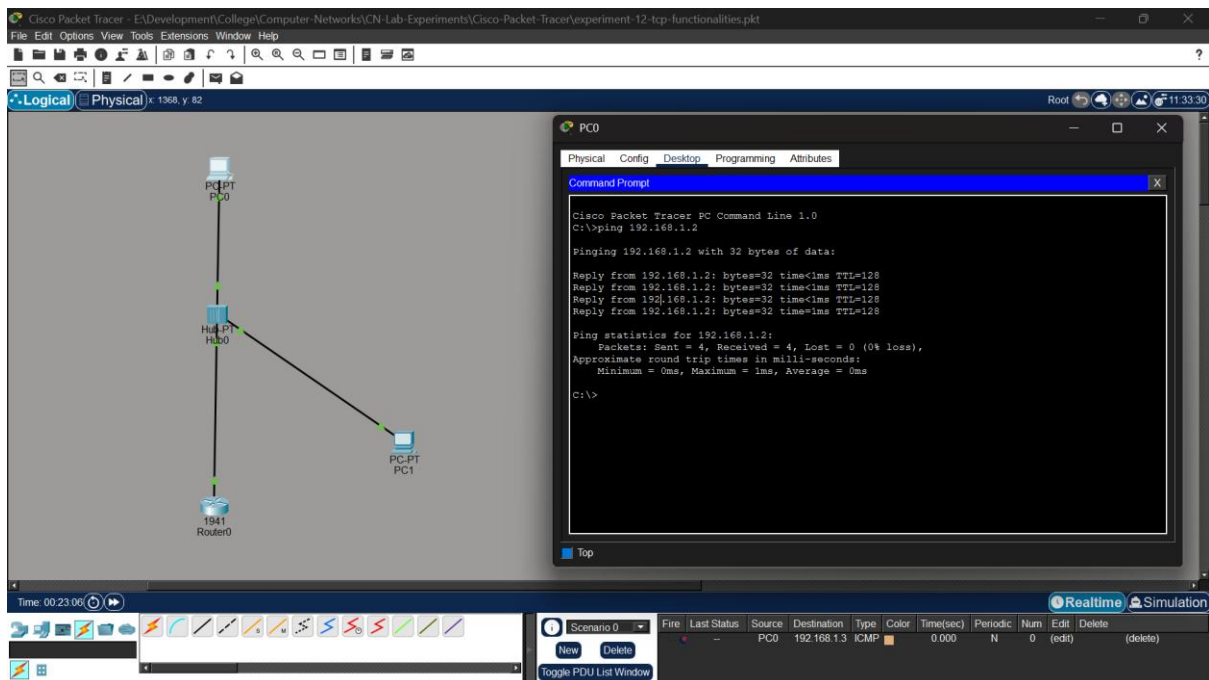
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
```

Experiment-11-Computer-Lab Router to Router Communication Using Static Routing

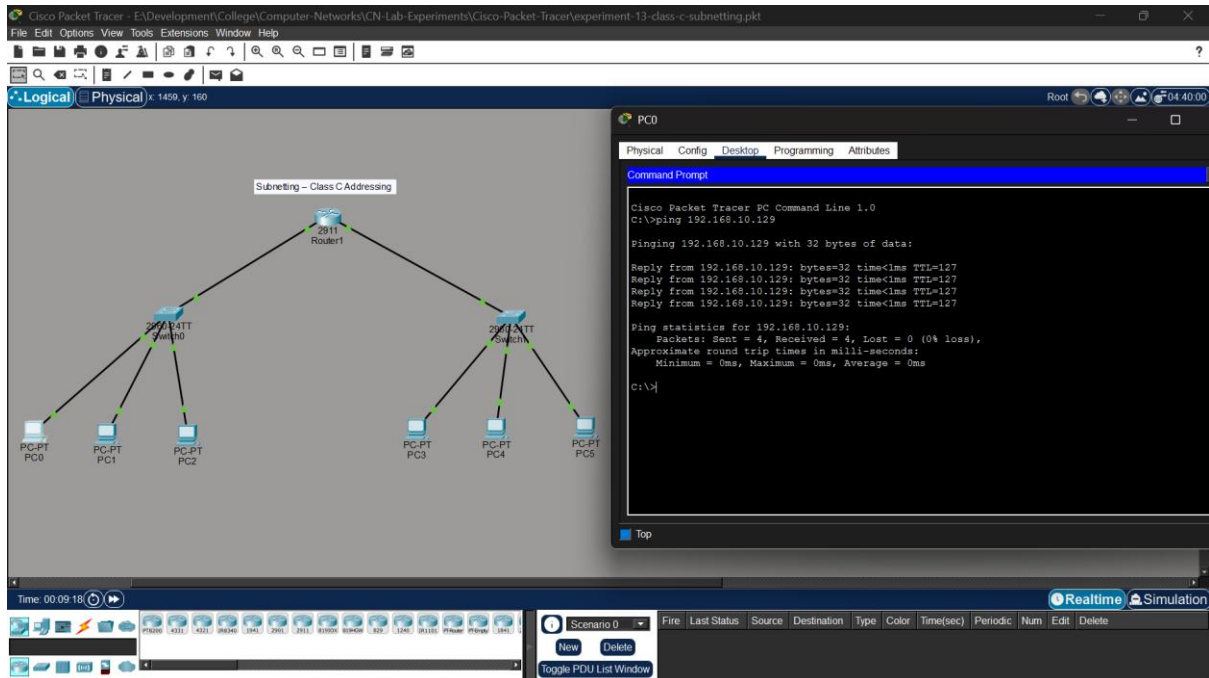




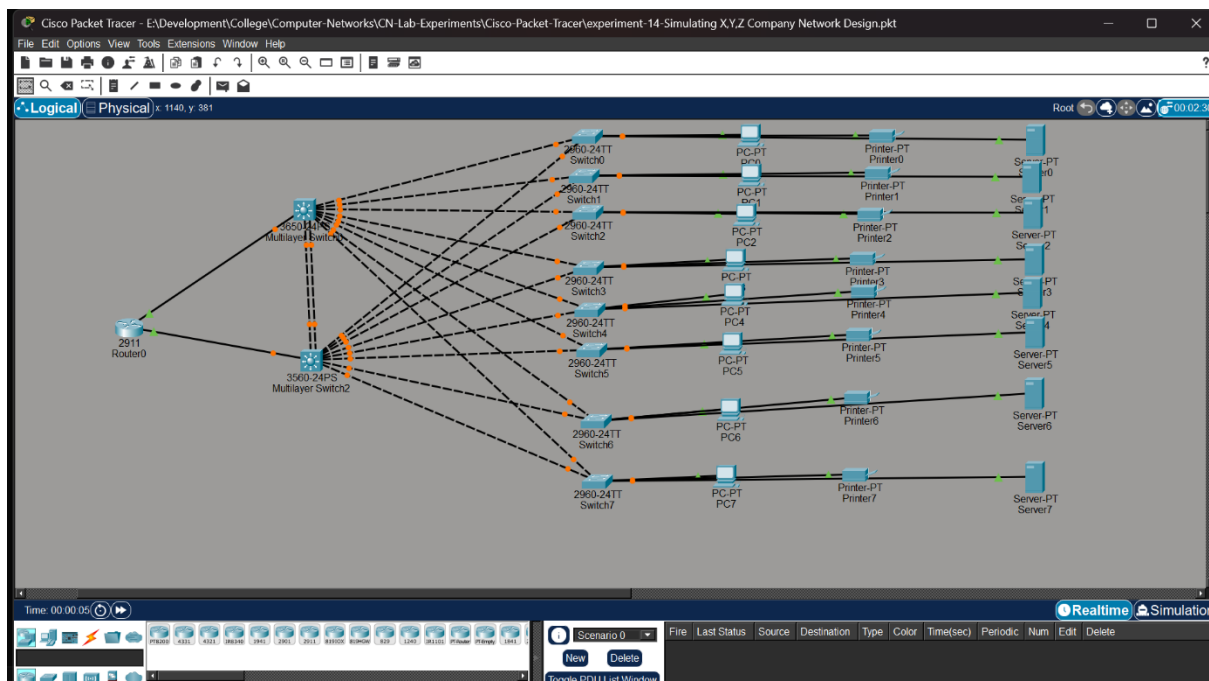
Experiment-12-Tcp-Functionalities



Experiment-13- Experiment-class-c-subnetting



Experiment-14- xyz-company-network-design



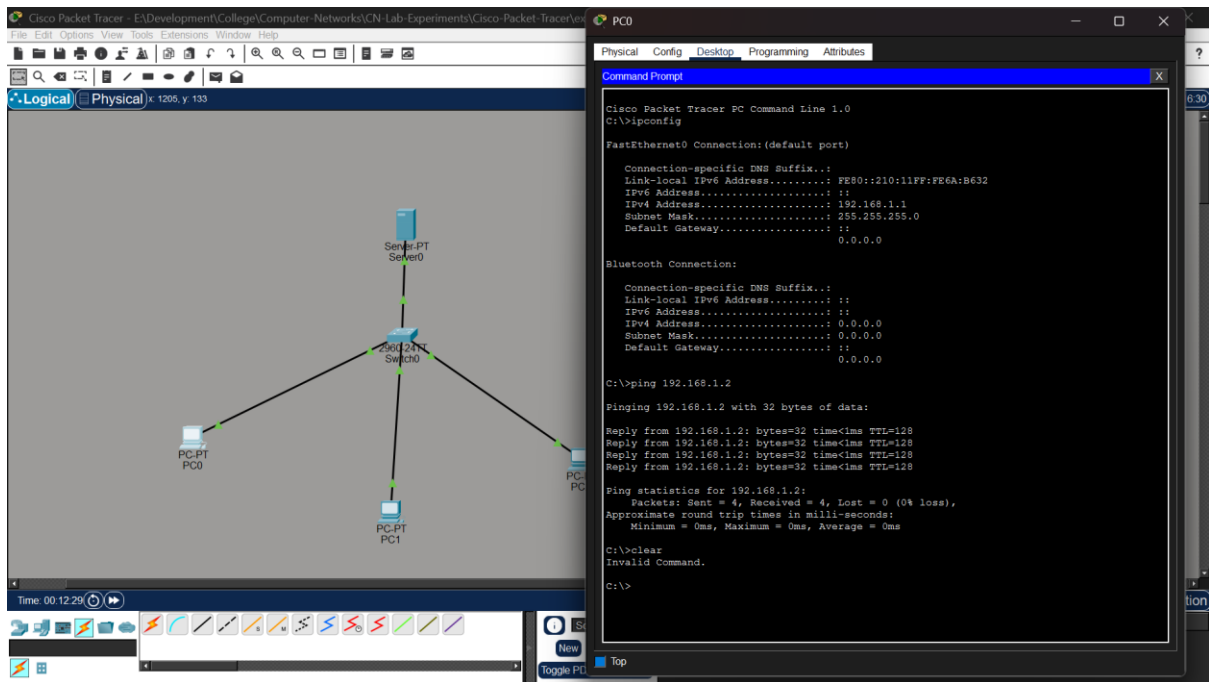
Experiment-15- DHCP Configuration

The image shows the Cisco Packet Tracer interface. On the left, a network topology is displayed in the Logical view. It consists of a central 2960 24-T Switch connected to three PC-PT devices (PC0, PC1, PC2) and a Server-PT. The time is 00:09:06. On the right, the configuration window for PC0 is open, showing the IP Configuration tab. The DHCP option is selected, and the configuration shows a successful DHCP request. The IPv6 Configuration tab is also visible, showing the Link Local Address FE80::210:11FF:FE0A:B032. The 802.1X Security tab is also visible, showing the Use 802.1X Security option selected.

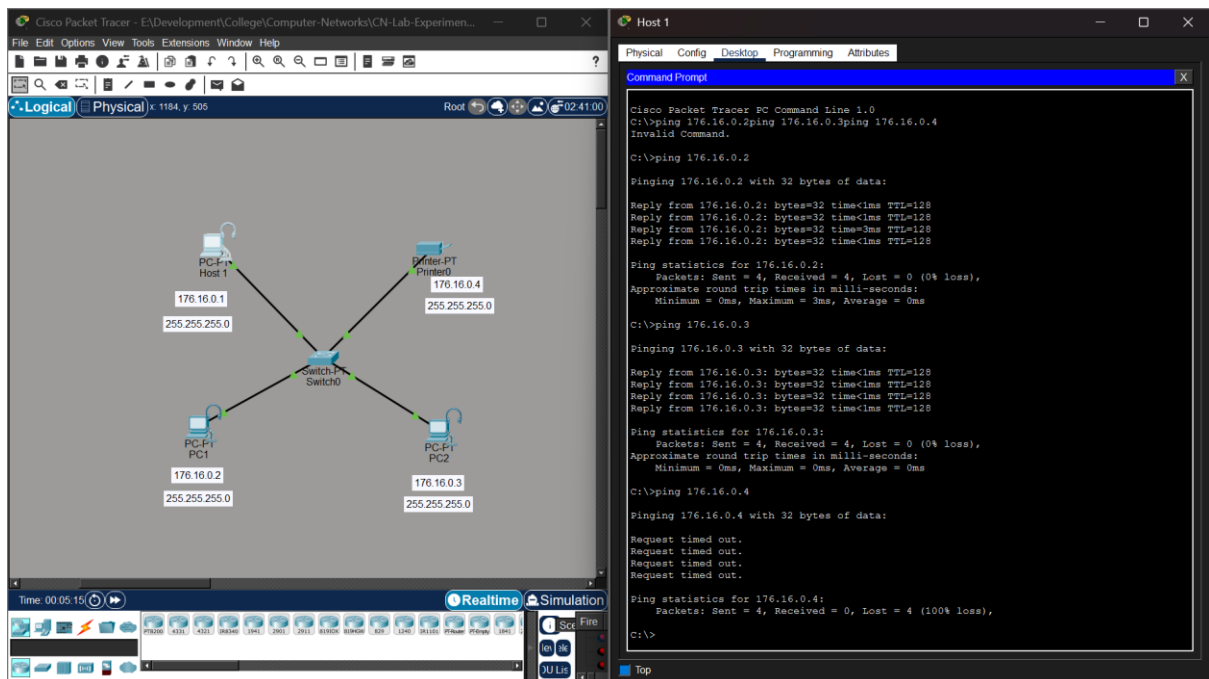
PC0 Configuration:

- Interface: FastEthernet0
- IP Configuration:
 - DHCP: ☒ DHCP, ☐ Static
 - Static IP Address: 192.168.1.1
 - Static Subnet Mask: 255.255.255.0
 - Static Default Gateway: 0.0.0.0
 - Static DNS Server: 0.0.0.0
- IPv6 Configuration:
 - Automatic: ☒ Automatic, ☐ Static
 - IPv6 Address: [Empty]
 - Link Local Address: FE80::210:11FF:FE0A:B032
 - Default Gateway: [Empty]
 - DNS Server: [Empty]
- 802.1X:
 - Use 802.1X Security: ☒
 - Authentication: MD5
 - Username: [Empty]
 - Password: [Empty]

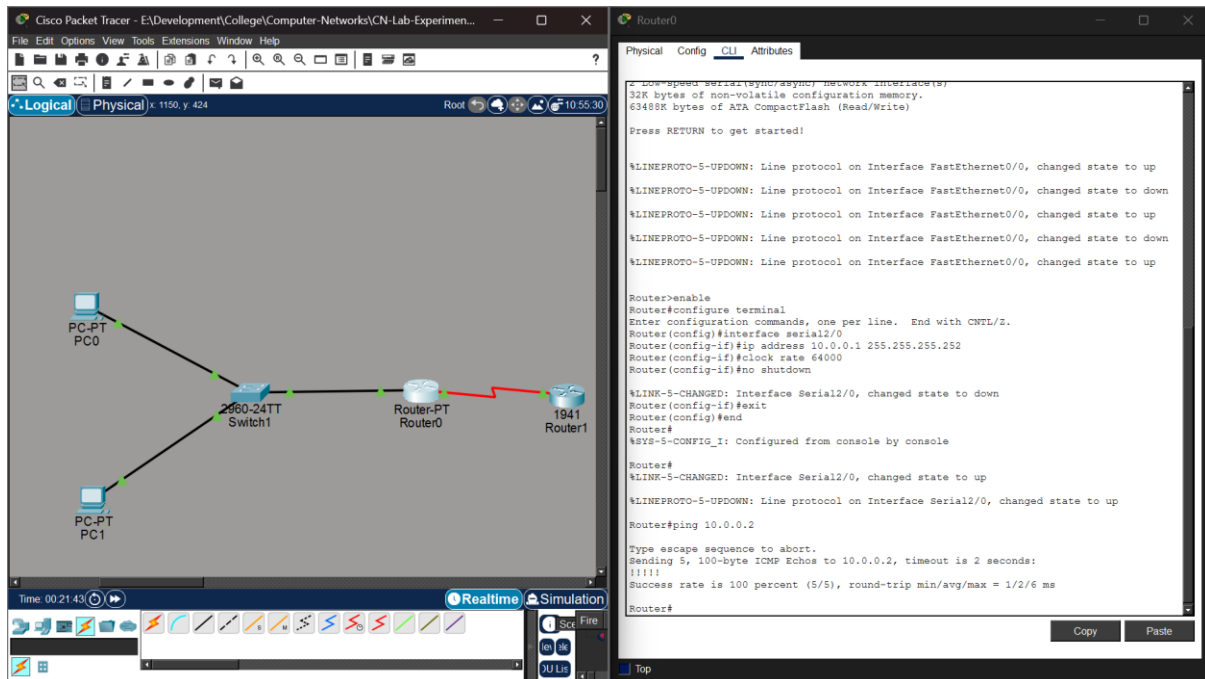
The image shows the same Cisco Packet Tracer interface, but now in the Realtime simulation mode. The network topology remains the same, with the 2960 24-T Switch connected to PC0, PC1, PC2, and the Server-PT. The time is 00:08:46. The bottom status bar shows "Realtime" and "Simulation" modes. The bottom right corner displays the time 04:24:30.



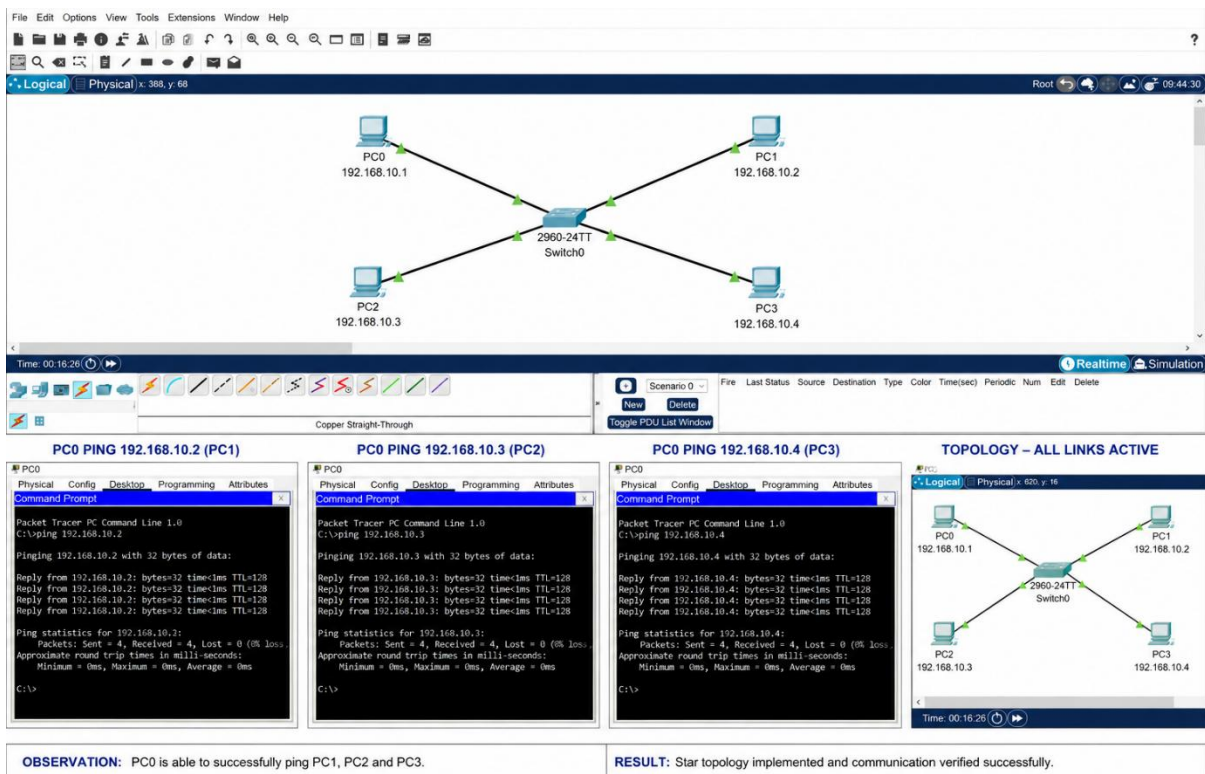
Experiment-16- Fire Wall



Experiment 17 – Firewall Configuration and Network Connectivity



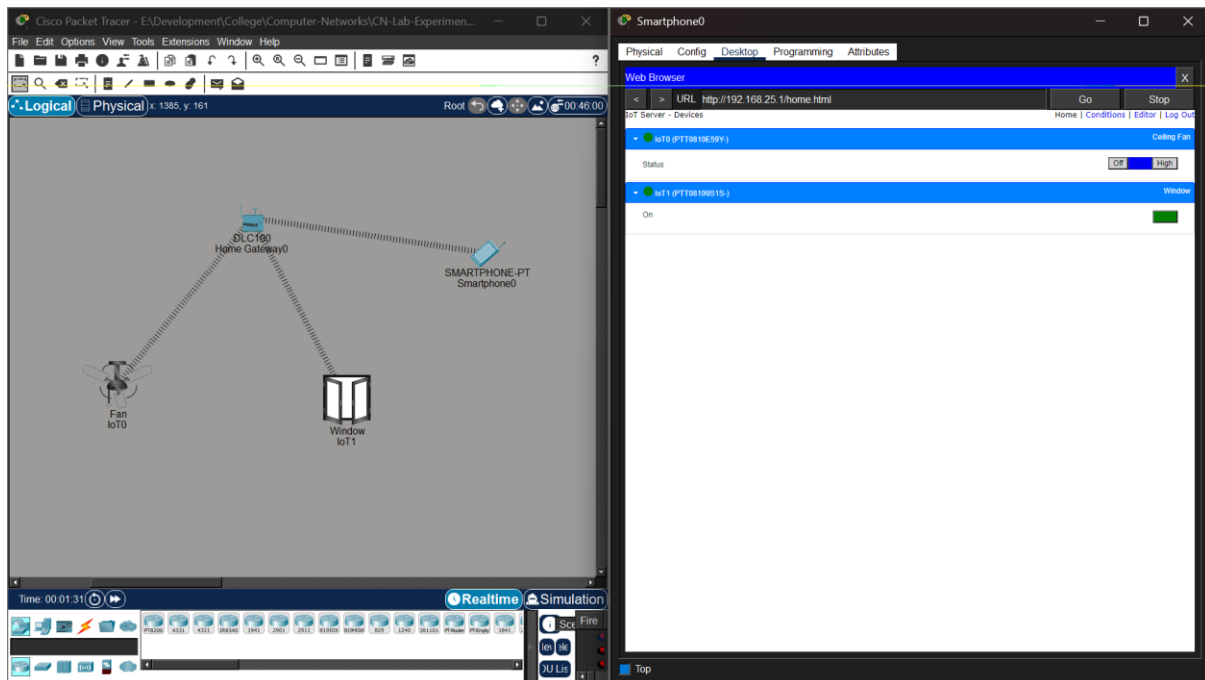
Experiment-18-Multimedia Network



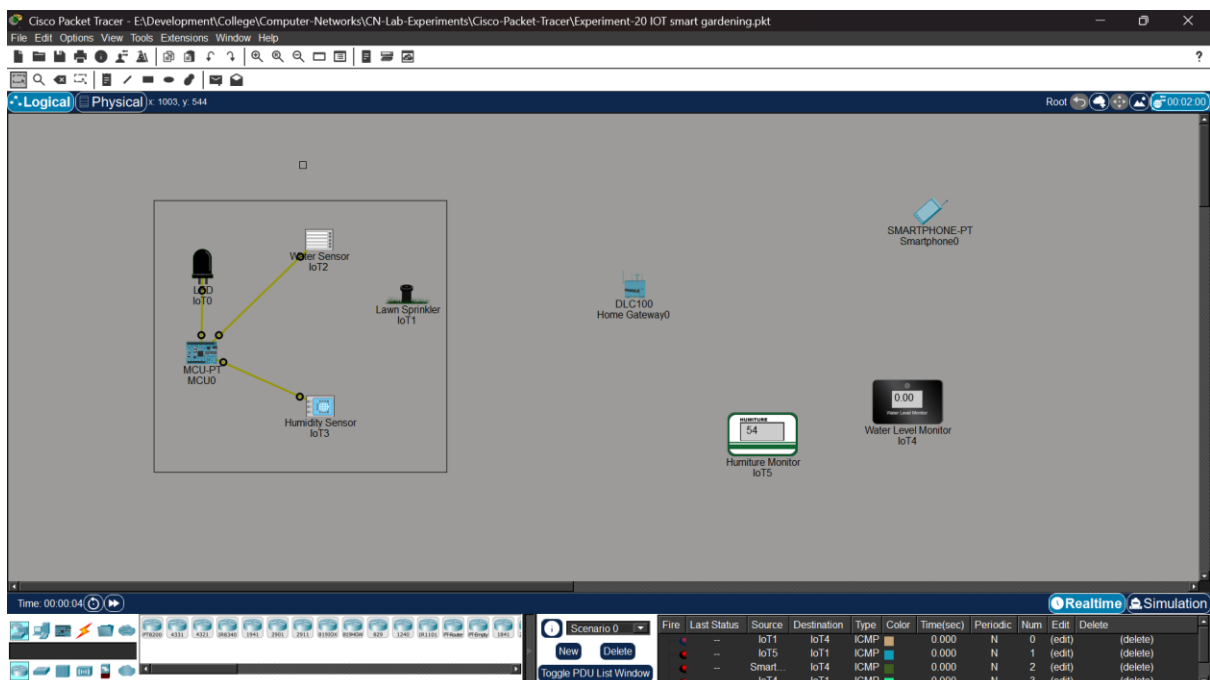
OBSERVATION: PC0 is able to successfully ping PC1, PC2 and PC3.

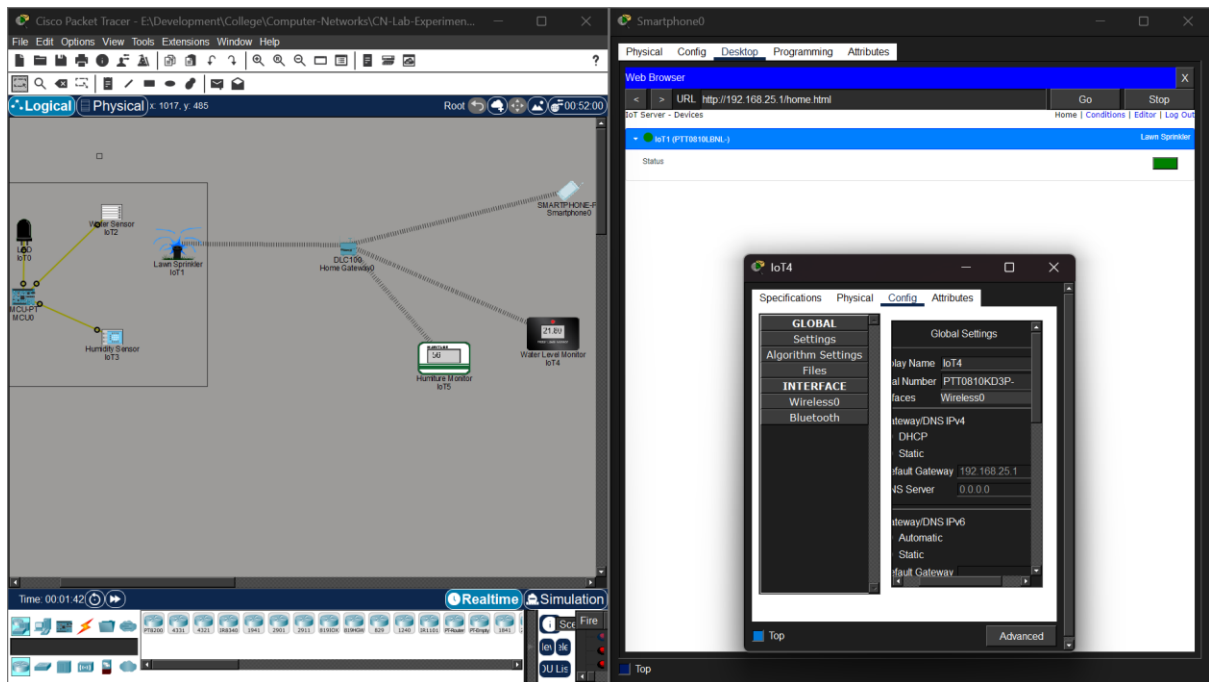
RESULT: Star topology implemented and communication verified successfully.

Experiment-19- IoT based smart home applications

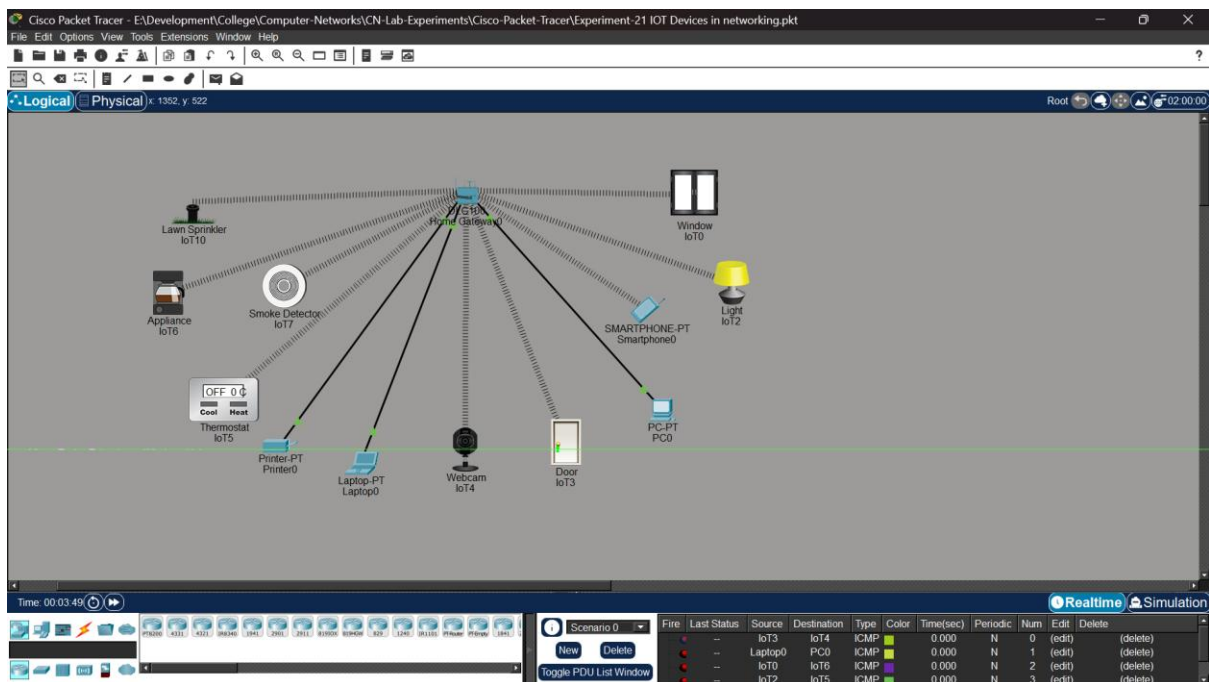


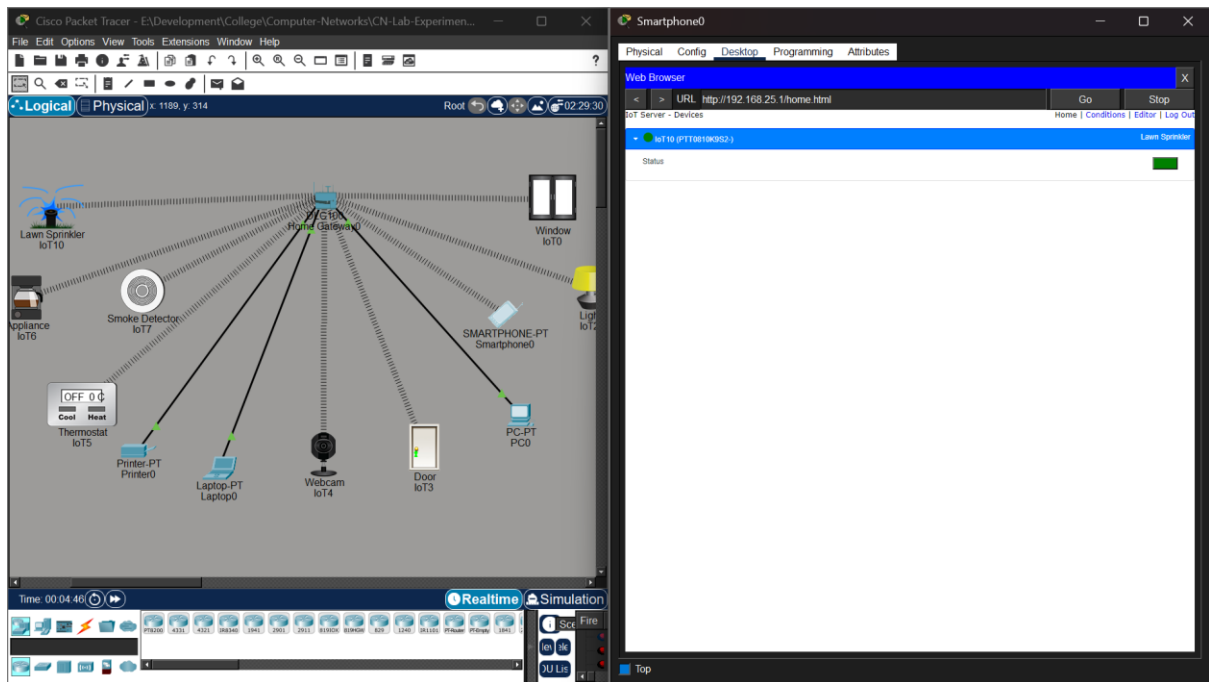
Experiment-20- IoT based smart home applications





Experiment-21- IoT devices in networking





Experiment-22- IoT based AAA Local and Server

Cisco Packet Tracer - E:\Development\College\Computer-Networks\CN-Lab-Experiments\Cisco-Packet-Tracer\Experiment-22-IoT based AAA Local and Server.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 393, y: 333

Root 03:10:30

Time: 00:49:58

Scenario 0

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful		Router0	PC0	ICMP		0.000	N	0	(edit)	(delete)
Successful		Router0	Server0	ICMP		0.000	N	1	(edit)	(delete)
Successful		PC0	Server0	ICMP		0.000	N	2	(edit)	(delete)

Toggle PDU List Window

Cisco Packet Tracer - E:\Development\College\Computer-Networks\CN-Lab-Experiments\Cisco-Packet-Tracer\Experiment-22-IoT based AAA Local and Server.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1503, y: 543

Root 03:02:30

Time: 00:49:49.853

Scenario 0

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful		Router0	PC0	ICMP		0.000	N	0	(edit)	(delete)
Successful		Router0	Server0	ICMP		0.000	N	1	(edit)	(delete)
Successful		PC0	Server0	ICMP		0.000	N	2	(edit)	(delete)

Toggle PDU List Window

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.002	Router0
	0.002	Switch0
	0.002	Switch0
	0.002	Switch0
	0.002	PC0
	0.003	Switch0
	0.003	Switch0
	0.003	PC0
	0.004	Switch0
	0.004	Server0
	0.004	Switch0
	0.005	Server0
	0.005	Switch0
	0.006	Switch0

Reset Simulation

Constant Delay

Captured to...

0.006 s

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT-TCP, LACP, LLDP, MDIOBUS, NetFlow, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Protocol, RADIUS, REP, RFP, RFPng, RTP, SCCP, SMTP, SNMP, SSH, SIP, Syslog, TACACS, TFTP, TFTP, Telnet, UDP, USB, VTP

Edit Filters

Show All/None

Event List

Realtime

Simulation

Experiment-23- TLP header analysis using Wire shark- TCP and UDP

The image shows a Wireshark capture of a TCP connection. The packet list on the left shows a series of packets, including a SYN exchange and a data packet. The selected packet (No. 11792) is a TCP segment with sequence number 2825, acknowledgment number 1732, and length 1225. The packet details pane on the right shows the structure of the TCP segment, including the source and destination ports, sequence number, acknowledgment number, and the data payload. The packet bytes pane on the right shows the raw data of the packet, including the TCP header and the application data.

No.	Time	Source	Destination	Protocol	Length	Info
11586	33.579227100	142.251.222.170	172.18.109.19	TLSv1.3	1078	Application Data, Application Data
11588	33.624620000	172.18.109.19	142.251.222.170	TCP	54	51435 → 443 [ACK] Seq=1870 Ack=18630 Win=64256 Len=0
11597	33.860895100	172.64.148.235	172.18.109.19	TLSv1.2	82	Application Data
11602	33.861752600	172.18.109.19	172.64.148.235	TLSv1.2	82	Application Data
11605	33.868520800	172.64.148.235	172.18.109.19	TCP	58	443 → 53388 [ACK] Seq=134 Ack=148 Win=19 Len=0
11732	34.338054700	172.18.109.19	142.251.222.131	TCP	66	55653 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
11752	34.351869000	142.251.222.131	172.18.109.19	TCP	66	443 → 55653 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
11755	34.352007300	172.18.109.19	142.251.222.131	TCP	54	55653 → 443 [ACK] Seq=0 Ack=1 Win=65280 Len=0
11758	34.353129200	172.18.109.19	142.251.222.131	TLSv1.3	1785	Client Hello (SNI=ssl.gstatic.com)
11766	34.379260100	142.251.222.131	172.18.109.19	TCP	54	443 → 55653 [ACK] Seq=1 Ack=1413 Win=268288 Len=0
11769	34.379263400	142.251.222.131	172.18.109.19	TCP	54	[TCP Dup ACK 11766#1] 443 → 55653 [ACK] Seq=1 Ack=1413 Win=268288 Len=0
11768	34.379262700	142.251.222.131	172.18.109.19	TCP	54	443 → 55653 [ACK] Seq=1 Ack=1732 Win=268032 Len=0
11791	34.388330800	142.251.222.131	172.18.109.19	TLSv1.3	2878	Server Hello, Change Cipher Spec
11792	34.388332100	142.251.222.131	172.18.109.19	TLSv1.3	1279	Application Data
11793	34.388466100	172.18.109.19	142.251.222.131	TCP	54	55653 → 443 [ACK] Seq=1732 Ack=4050 Win=65280 Len=0
11798	34.391706900	172.18.109.19	142.251.222.131	TLSv1.3	128	Change Cipher Spec, Application Data

Frame 11792: Packet, 1279 bytes on wire (10232 bits), 1279 bytes captured (10232 bits) on interface...
Ethernet II, Src: Cisco_66:9c:4b (00:26:0b:66:9c:4b), Dst: Intel_I:ef:4b (2c:6d:c1:1f:ef:4b)
Internet Protocol Version 4, Src: 142.251.222.131, Dst: 172.18.109.19
Transmission Control Protocol, Src Port: 443, Dst Port: 55653, Seq: 2825, Ack: 1732, Len: 1225
Source Port: 443
Destination Port: 55653
[Stream index: 13]
[Stream Packet Number: 9]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 1225]
Sequence Number: 2825 (relative sequence number)
Sequence Number (raw): 2555334328
[Next Sequence Number: 4050 (relative sequence number)]
Acknowledgment Number: 1732 (relative ack number)
Acknowledgment Number (raw): 3766573883
8188... Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)

The image shows a Wireshark capture of a UDP connection. The packet list on the left shows a single packet (No. 11795) with source port 58669 and destination port 443. The packet details pane on the right shows the structure of the UDP segment, including the source and destination ports, length, and the data payload. The packet bytes pane on the right shows the raw data of the packet, including the UDP header and the application data.

No.	Time	Source	Destination	Protocol	Length	Info
11782	34.388974600	172.18.109.19	142.251.150.119	UDP	74	59562 → 443 Len=32
11783	34.381423300	172.18.109.19	172.217.24.202	QUIC	73	Protected Payload (KP0), DCID=ee8a49e3a134ee8
11784	34.381535900	172.18.109.19	142.251.222.131	QUIC	115	Handshake, DCID=fd3fc287a7301c60
11785	34.381604500	172.18.109.19	142.250.205.106	QUIC	74	Protected Payload (KP0), DCID=ab16d3910c2653b1
11786	34.381705100	172.18.109.19	142.251.222.131	QUIC	75	Protected Payload (KP0), DCID=fd3fc287a7301c60
11787	34.388321600	142.251.222.131	172.18.109.19	QUIC	570	Protected Payload (KP0)
11788	34.388323300	142.251.222.131	172.18.109.19	QUIC	64	Protected Payload (KP0)
11789	34.388324600	142.251.222.131	172.18.109.19	QUIC	128	Protected Payload (KP0)
11790	34.388325400	142.251.222.131	172.18.109.19	QUIC	64	Protected Payload (KP0)
11794	34.388885000	142.251.150.119	172.18.109.19	UDP	67	443 → 59562 Len=25
11795	34.390878200	172.18.109.19	142.251.222.131	QUIC	79	Protected Payload (KP0), DCID=fd3fc287a7301c60
11796	34.390874800	172.18.109.19	142.251.222.131	QUIC	75	Protected Payload (KP0), DCID=fd3fc287a7301c60
11797	34.390972000	172.18.109.19	142.251.222.131	QUIC	79	Protected Payload (KP0), DCID=fd3fc287a7301c60
11799	34.393102900	172.217.24.138	172.18.109.19	QUIC	204	Protected Payload (KP0)
11800	34.393105000	142.250.205.106	172.18.109.19	QUIC	1055	Protected Payload (KP0)
11801	34.393106000	142.250.205.106	172.18.109.19	QUIC	86	Protected Payload (KP0)

Frame 11795: Packet, 79 bytes on wire (632 bits), 79 bytes captured (632 bits) on interface...
Ethernet II, Src: Intel_I:ef:4b (2c:6d:c1:1f:ef:4b), Dst: Cisco_66:9c:4b (00:26:0b:66:9c:4b)
Internet Protocol Version 4, Src: 172.18.109.19, Dst: 142.251.222.131
User Datagram Protocol, Src Port: 58669, Dst Port: 443
Source Port: 58669
Destination Port: 443
Length: 45
Checksum: 0x86e3 [unverified]
[Checksum Status: Unverified]
[Stream index: 164]
[Stream Packet Number: 25]
[Timestamps]
UDP payload (37 bytes)
QUIC IETF

Experiment-24 - Network Layer Protocol Header Analysis using Wireshark – SMTP and ICMP

The screenshot shows a Wireshark capture of ICMP Echo (ping) requests and replies. The packet list on the left shows several ICMP Echo (ping) requests and replies. The selected packet (10010) is an ICMP Echo (ping) request. The packet details pane on the right shows the ICMP Echo (ping) request details, including the Type (Echo (ping) request (8)), Code (0), Checksum (0x4d57), Identifier (BE: 1 (0x0001), LE: 256 (0x0100)), Sequence Number (BE: 4 (0x0004), LE: 1024 (0x0400)), and Data (32 bytes). The packet bytes pane on the right shows the raw data of the ICMP Echo (ping) request, including the Type (8), Code (0), Checksum (0x4d57), Identifier (1), Sequence Number (4), and Data (32 bytes).

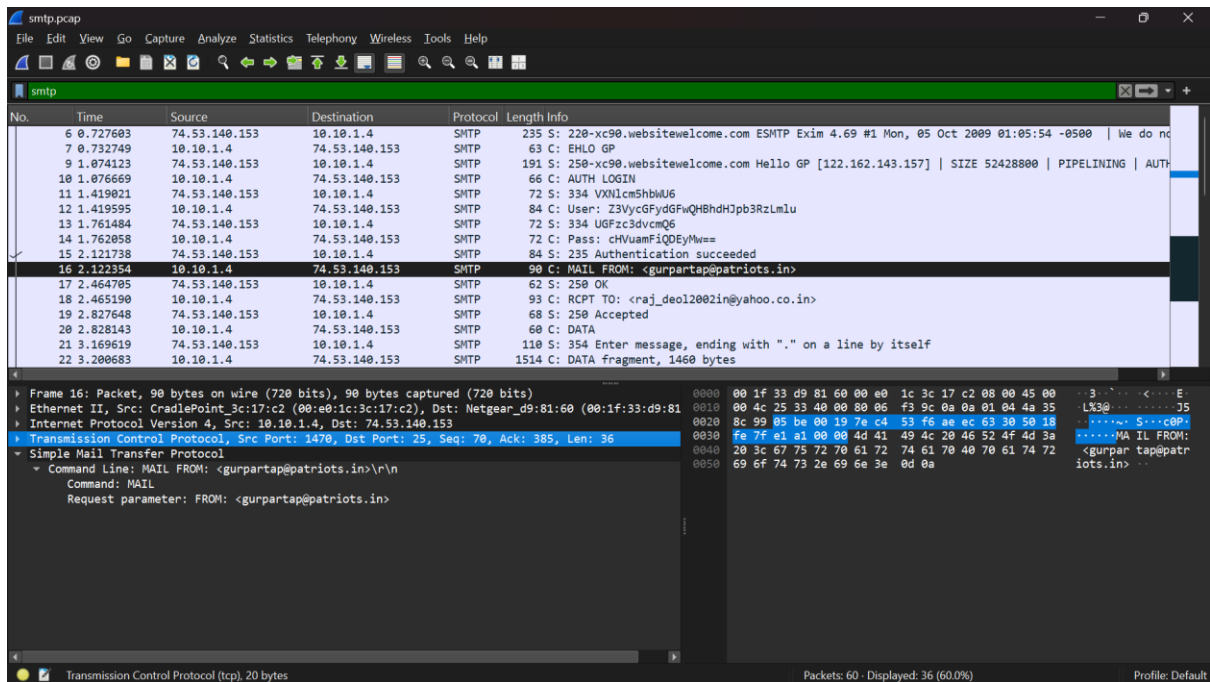
No.	Time	Source	Destination	Protocol	Length	Info
9166	20.034812100	172.25.42.125	172.217.24.206	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 9173)
9173	20.085888000	172.217.24.206	172.25.42.125	ICMP	78	Echo (ping) reply id=0x0001, seq=1/256, ttl=117 (request in 9166)
9451	21.044649000	172.25.42.125	172.217.24.206	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 9453)
9453	21.075387000	172.217.24.206	172.25.42.125	ICMP	78	Echo (ping) reply id=0x0001, seq=2/512, ttl=117 (request in 9451)
9756	22.056933000	172.25.42.125	172.217.24.206	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 9762)
9762	22.067441600	172.217.24.206	172.25.42.125	ICMP	78	Echo (ping) reply id=0x0001, seq=3/768, ttl=117 (request in 9756)
10010	23.063652000	172.25.42.125	172.217.24.206	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 10012)
10012	23.094599200	172.217.24.206	172.25.42.125	ICMP	78	Echo (ping) reply id=0x0001, seq=4/1024, ttl=117 (request in 10010)

Frame 10010: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF{...} Ethernet II, Src: Intel_lf:ef:4b (2c:6d:c1:1f:ef:4b), Dst: Cisco_gd:c0:e7 (24:d5:e4:6d:c0:e7) Internet Protocol Version 4, Src: 172.25.42.125, Dst: 172.217.24.206 Internet Control Message Protocol (ICMP) Type: Echo (ping) request (8) Code: 0 Checksum: 0x4d57 [correct] [Checksum Status: Good] Identifier (BE): 1 (0x0001) Identifier (LE): 256 (0x0100) Sequence Number (BE): 4 (0x0004) Sequence Number (LE): 1024 (0x0400) [Response frame: 10012] Data (32 bytes)

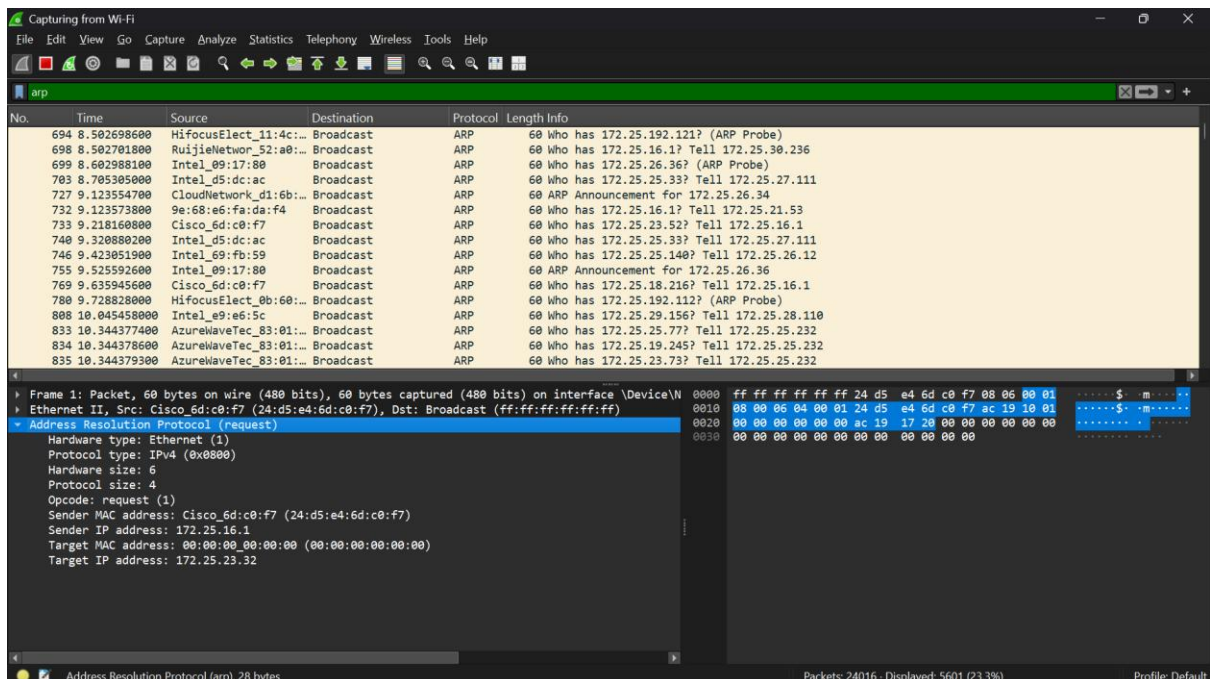
The screenshot shows a Wireshark capture of an SMTP session. The packet list on the left shows several SMTP packets. The selected packet (11) is an SMTP packet. The packet details pane on the right shows the SMTP packet details, including the Source Port (25), Destination Port (1470), Stream Index (0), Stream Packet Number (9), Conversation completeness (Complete, WITH_DATA (31)), TCP Segment Len (18), Sequence Number (319), Sequence Number (raw) (2934727486), Next Sequence Number (337), Acknowledgment Number (22), Acknowledgment number (raw) (2126795718), 0101 ... = Header Length: 20 bytes (5), and Flags (0x018 (PSH, ACK)). The packet bytes pane on the right shows the raw data of the SMTP packet, including the Source Port (25), Destination Port (1470), Stream Index (0), Stream Packet Number (9), Conversation completeness (Complete, WITH_DATA (31)), TCP Segment Len (18), Sequence Number (319), Sequence Number (raw) (2934727486), Next Sequence Number (337), Acknowledgment Number (22), Acknowledgment number (raw) (2126795718), 0101 ... = Header Length: 20 bytes (5), and Flags (0x018 (PSH, ACK)).

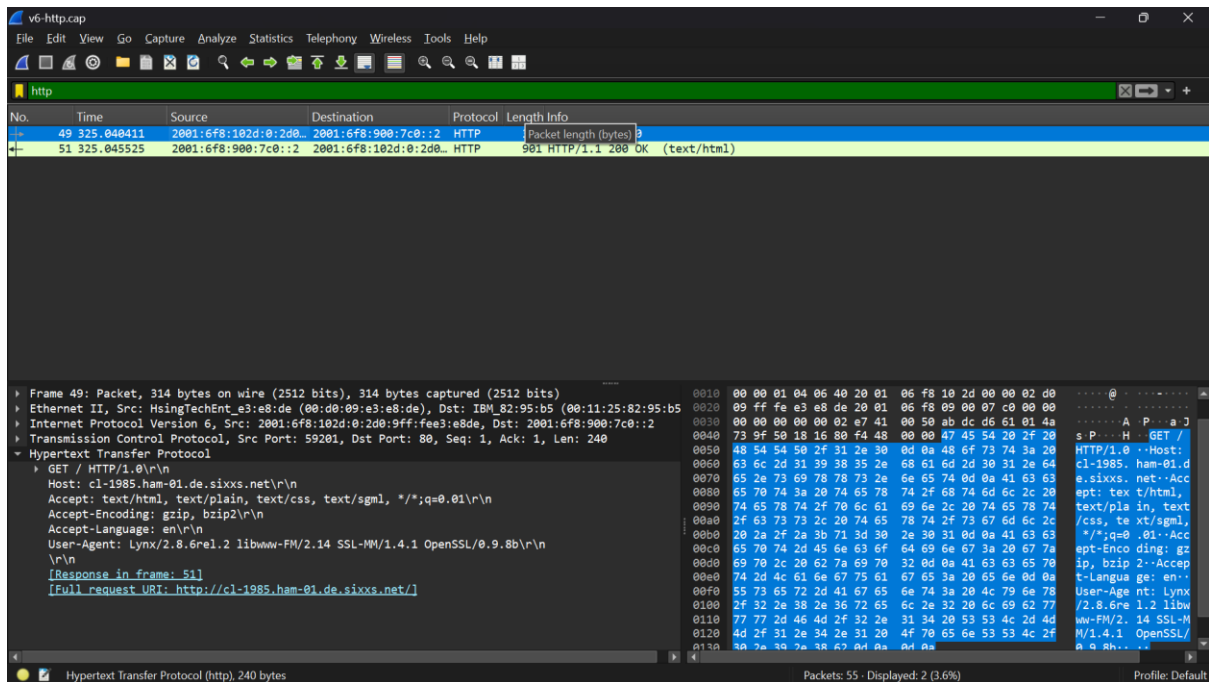
No.	Time	Source	Destination	Protocol	Length	Info
6	0.727693	74.53.140.153	10.10.1.4	SMTP	235	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 We do not
7	0.732749	10.10.1.4	74.53.140.153	SMTP	63	C: EHLO GP
9	1.074123	74.53.140.153	10.10.1.4	SMTP	191	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZE 52428800 PIPELINING AUTH
10	1.076669	10.10.1.4	74.53.140.153	SMTP	66	C: AUTH LOGIN
11	1.419821	74.53.140.153	10.10.1.4	SMTP	72	S: 334 VXNlcmShbmU6
12	1.419595	10.10.1.4	74.53.140.153	SMTP	84	C: User: Z3VycG9yZG90H3b3R2Zmlu
13	1.761484	74.53.140.153	10.10.1.4	SMTP	72	S: 334 UGFzc3dvcmQ6
14	1.762058	10.10.1.4	74.53.140.153	SMTP	72	C: Pass: cHVuamF1QDEyMw==
15	2.121738	74.53.140.153	10.10.1.4	SMTP	84	S: 235 Authentication succeeded
16	2.122354	10.10.1.4	74.53.140.153	SMTP	90	C: MAIL FROM: <gurpartap@patriots.in>
17	2.464705	74.53.140.153	10.10.1.4	SMTP	62	S: 250 OK
18	2.465190	10.10.1.4	74.53.140.153	SMTP	93	C: RCPT TO: <raj_deol2002@yahoo.co.in>
19	2.827648	74.53.140.153	10.10.1.4	SMTP	68	S: 250 Accepted
20	2.828143	10.10.1.4	74.53.140.153	SMTP	60	C: DATA
21	3.169619	74.53.140.153	10.10.1.4	SMTP	110	S: 354 Enter message, ending with "." on a line by itself
22	3.200683	10.10.1.4	74.53.140.153	SMTP	1514	C: DATA fragment, 1460 bytes

Frame 11: Packet, 72 bytes on wire (576 bits), 72 bytes captured (576 bits) Ethernet II, Src: Netgear_d9:81:60 (00:1f:33:d9:81:60), Dst: CradlePoint_3c:17:c2 (00:e0:1c:3c:17:c2) Internet Protocol Version 4, Src: 74.53.140.153, Dst: 10.10.1.4 Transmission Control Protocol, Src Port: 25, Dst Port: 1470, Seq: 319, Ack: 22, Len: 18 Source Port: 25 Destination Port: 1470 [Stream index: 0] [Stream Packet Number: 9] [Conversation completeness: Complete, WITH_DATA (31)] [TCP Segment Len: 18] Sequence Number: 319 (relative sequence number) Sequence Number (raw): 2934727486 [Next Sequence Number: 337 (relative sequence number)] Acknowledgment Number: 22 (relative ack number) Acknowledgment number (raw): 2126795718 0101 ... = Header Length: 20 bytes (5) Flags: 0x018 (PSH, ACK)

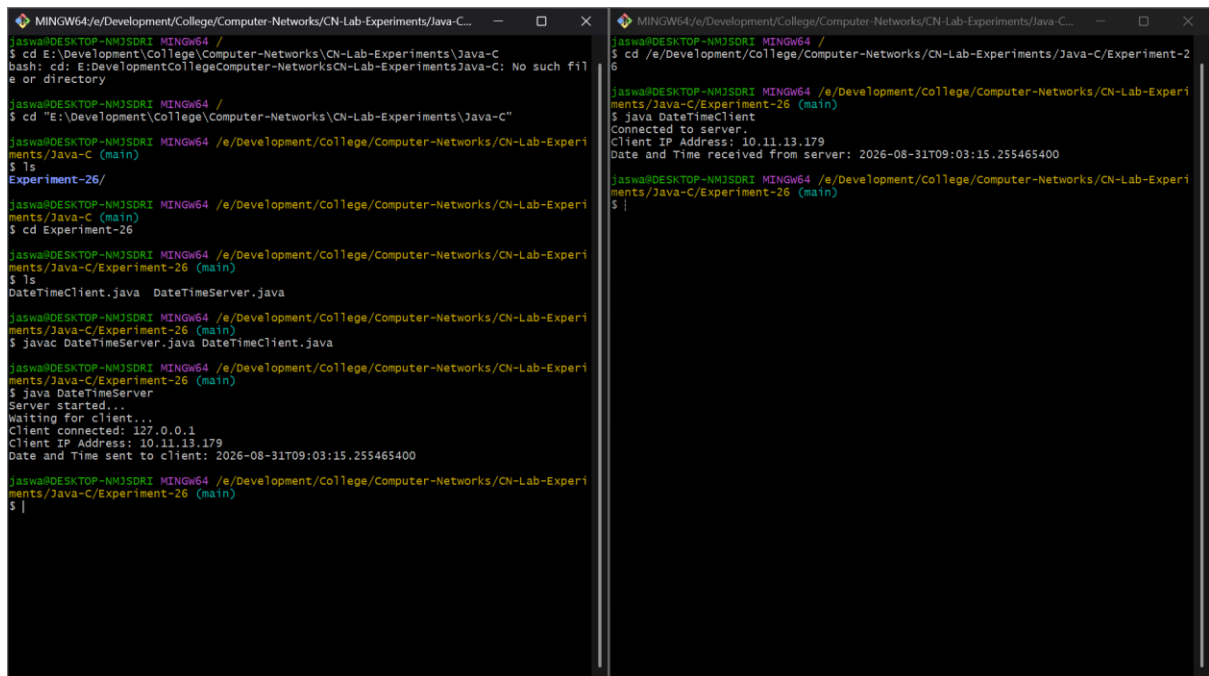


Experiment 25 – Network Layer Protocol Header Analysis using Wireshark – ARP and HTTP

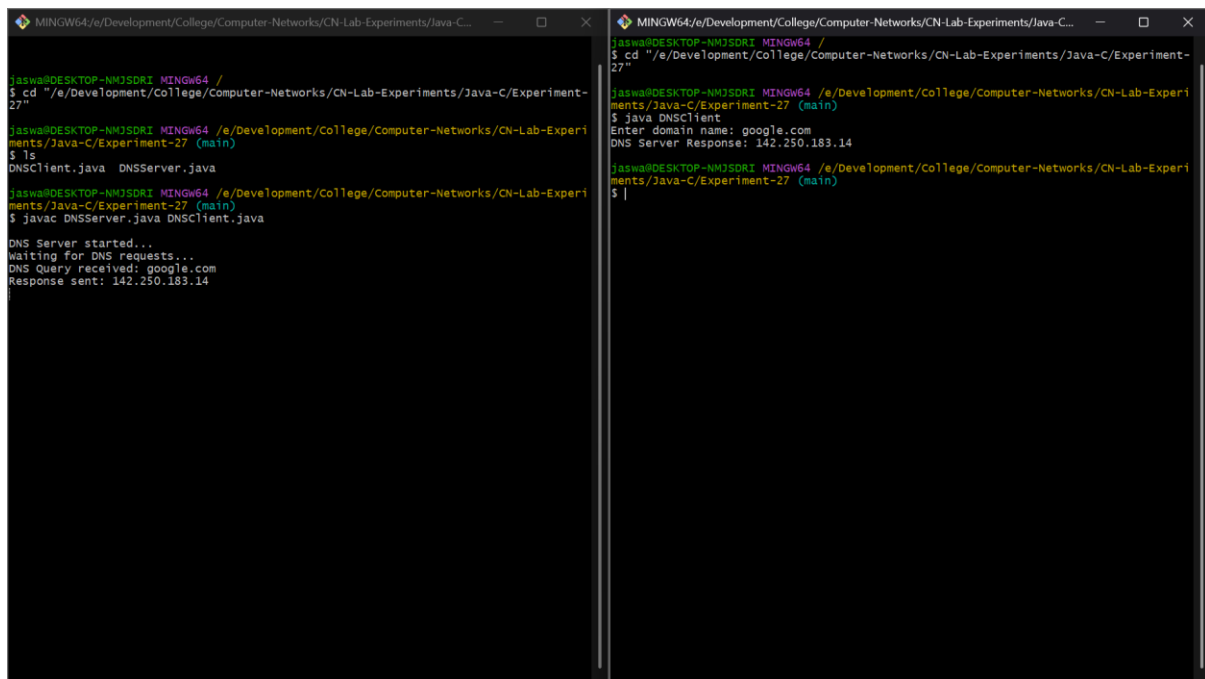




Experiment 26 – Date and Time Display from Client to Server using TCP Socket



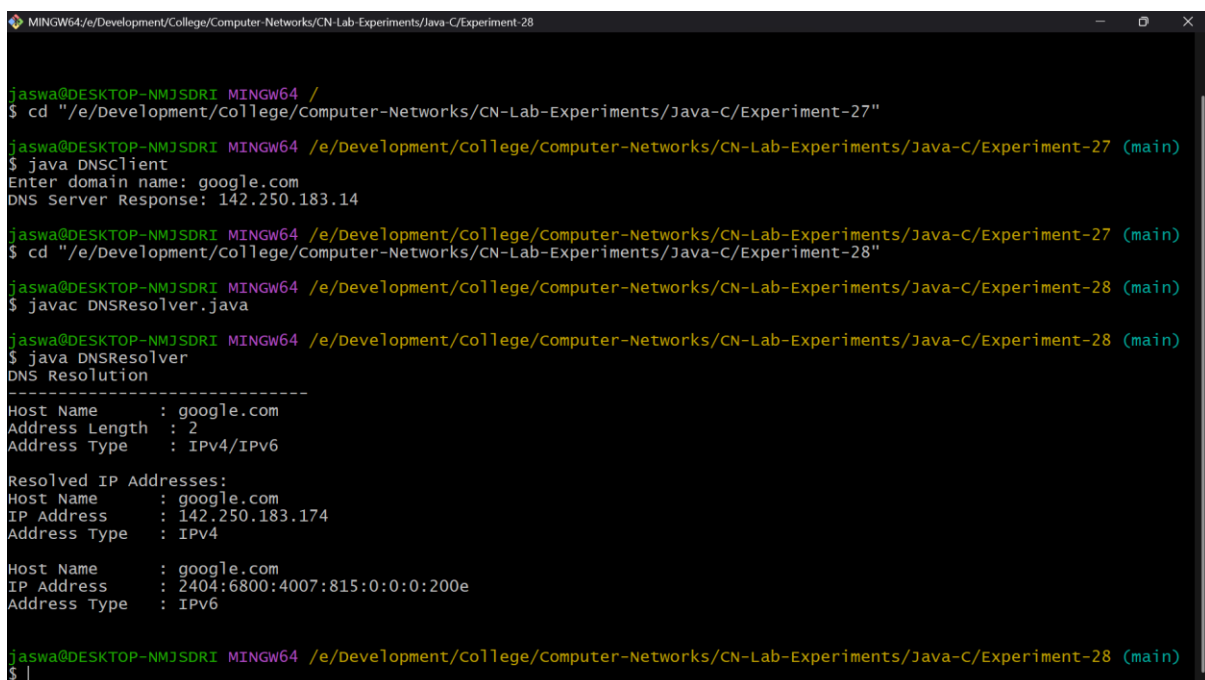
Experiment 27 – Implementation of DNS Server and Client using UDP Socket



```
MINGW64/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C...
jaswa@DESKTOP-NM3SDRI MINGW64 /
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-27"
$ ls
DNSClient.java  DNSServer.java
$ javac DNSClient.java DNSServer.java
DNS Server started...
Waiting for DNS requests...
DNS Query received: google.com
Response sent: 142.250.183.14

MINGW64/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-27
jaswa@DESKTOP-NM3SDRI MINGW64 /
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-27"
$ java DNSClient
Enter domain name: google.com
DNS Server Response: 142.250.183.14
```

Experiment 28 – DNS Client to Resolve a Given Hostname using Java



```
MINGW64/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-28
jaswa@DESKTOP-NM3SDRI MINGW64 /
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-27"
$ java DNSClient
Enter domain name: google.com
DNS Server Response: 142.250.183.14

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-27 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-28"
$ javac DNSResolver.java

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-28 (main)
$ java DNSResolver
DNS Resolution
-----
Host Name      : google.com
Address Length : 2
Address Type   : IPV4/IPV6

Resolved IP Addresses:
Host Name      : google.com
IP Address     : 142.250.183.174
Address Type   : IPV4

Host Name      : google.com
IP Address     : 2404:6800:4007:815:0:0:0:200e
Address Type   : IPV6

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-28 (main)
$ |
```

Experiment 29 – TCP Echo Server and Client using Java

```
MINGW64/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C...
Address Length : 2
Address Type : IPv4/IPv6

Resolved IP Addresses:
Host Name : google.com
IP Address : 142.250.183.174
Address Type : IPv4

Host Name : google.com
IP Address : 2404:6800:4007:815:0:0:200e
Address Type : IPv6

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/
CN-Lab-Experiments/Java-C/Experiment-28 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-
C/Experiment-29"

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/
CN-Lab-Experiments/Java-C/Experiment-29 (main)
$ ls
EchoClient.java EchoServer.java

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/
CN-Lab-Experiments/Java-C/Experiment-29 (main)
$ javac EchoServer.java EchoClient.java

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/
CN-Lab-Experiments/Java-C/Experiment-29 (main)
$ java EchoServer
TCP Echo Server started...
waiting for client...
Client connected: 127.0.0.1
Received: java EchoClient
Received: Hello Server
Received: Computer Networks
Received: TCP Echo Test
Received: exit

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/
CN-Lab-Experiments/Java-C/Experiment-29 (main)
$ |

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-27 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-27"
$ ls
DNSClient.java DNSServer.java

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-27 (main)
$ javac DNSServer.java DNSClient.java

DNS Server started...
waiting for DNS requests...
DNS Query received: google.com
Response sent: 142.250.183.14

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-27 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-29"

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-29 (main)
$ java EchoClient
Connected to Echo Server.
Enter message: java EchoClient
Echo from server: java EchoClient
Enter message: Hello Server
Echo from server: Hello Server
Enter message: Computer Networks
Echo from server: Computer Networks
Enter message: TCP Echo Test
Echo from server: TCP Echo Test
Enter message: exit
Echo from server: exit

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-29 (main)
$ |
```

Experiment 30 – TCP Chat Client and Chat Server using Java

```
MINGW64/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C...
DNS Server started...
waiting for DNS requests...
DNS query received: google.com
Response sent: 142.250.183.14

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-27 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-29"

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-29 (main)
$ java EchoClient
Connected to Echo Server.
Enter message: java EchoClient
Echo from server: java EchoClient
Enter message: Hello Server
Echo from server: Hello Server
TCP Chat Server started...
waiting for client...
client connected: 127.0.0.1
Client: Hello Server
Server: Hello Client
Client: How are you?
Server: I am fine
Client ended the chat.

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-30 (main)
$ |

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-28 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-28"
$ javac DNSResolver.java

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-28 (main)
$ java DNSResolver
DNS Resolution
-----
Host Name : google.com
Address Length : 2
Address Type : IPv4/IPv6

Resolved IP Addresses:
Host Name : google.com
IP Address : 142.250.183.174
Address Type : IPv4

Host Name : google.com
IP Address : 2404:6800:4007:815:0:0:200e
Address Type : IPv6

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-28 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-29"

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-29 (main)
$ java ChatServer
Connected to Chat Server.
Client: Hello Server
Server: Hello Client
Client: How are you?
Server: I am fine
Client: exit
Chat terminated.

jaswa@DESKTOP-NMJSORI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-30 (main)
$ |
```

Experiment 31 – Implementation of ARP Protocols in Java

```
MINGW64:/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-31
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-30 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-31"
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-31 (main)
$ javac ARPImplementation.java
java ARPImplementation
ARP Protocol Implementation
-----
Host Name      : DESKTOP-NM1SDRI
IP Address     : 10.11.13.179
MAC Address    : 2C:6D:C1:1F:EF:4B

ARP address resolution completed successfully.
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-31 (main)
$
```

Experiment 32 – Implementation of Bit Stuffing Mechanism using Java

```
MINGW64:/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-30 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-31"
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-31 (main)
$ javac ARPImplementation.java
java ARPImplementation
ARP Protocol Implementation
-----
Host Name      : DESKTOP-NM1SDRI
IP Address     : 10.11.13.179
MAC Address    : 2C:6D:C1:1F:EF:4B

ARP address resolution completed successfully.
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-31 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32"
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$ gcc BitStuffing.c -o BitStuffing
bash: gcc: command not found
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$ ls
Bitstuffing.c
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$ rm BitStuffing.c
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$ javac BitStuffing.java
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$ java BitStuffing
Bit Stuffing Mechanism
-----
Input Data   : 011111101111110
Flag        : 01111110
Stuffed Data : 011111100111111010111110100111110
Rule         : Insert 0 after five consecutive 1s
jaswa@DESKTOP-NM1SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$
```

Experiment 33 – TCP File Transfer using Java

```
MINGW64/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C...
[loading /modules/jdk.hotspot.agent/module-info.class]
[loading /modules/jdk.jartool/module-info.class]
[loading /modules/jdk.net/module-info.class]
[loading /modules/jdk.compiler/module-info.class]
[loading /modules/jdk.management.agent/module-info.class]
[loading /modules/jdk.jdi/module-info.class]
[loading /modules/java.sql.rowset/module-info.class]
[loading /modules/jdk.internal.opt/module-info.class]
[loading /modules/jdk.zipfs/module-info.class]
[search path for source files: ]
[search path for class files: C:\Users\jaswa\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.20.8-hotspot\lib\modules,.]
[total 191ms]

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-33 (main)
$ ls -la
total 13
drwxr-xr-x 1 jaswa 197609 0 Aug 31 10:09 ./
drwxr-xr-x 1 jaswa 197609 0 Aug 31 10:05 ../
-rw-r--r-- 1 jaswa 197609 1709 Aug 31 10:09 FileClient.class
-rw-r--r-- 1 jaswa 197609 1094 Aug 31 10:06 FileClient.java
-rw-r--r-- 1 jaswa 197609 0 Aug 31 10:05 FileServer.java
-rw-r--r-- 1 jaswa 197609 40 Aug 31 10:09 sample.txt

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-33 (main)
$ ls -l FileServer.java
-rw-r--r-- 1 jaswa 197609 1198 Aug 31 10:13 FileServer.java

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-33 (main)
$ javac FileServer.java

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-33 (main)
$ ls
FileClient.class FileClient.java FileServer.class FileServer.java sample.txt

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-33 (main)
$ java -cp . FileServer
TCP File Server started...
Waiting for client...
Client connected.
File transferred successfully.
File sent: sample.txt

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-33 (main)
$

bash: gcc: command not found

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$ ls
BitStuffing.c

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$ rm BitStuffing.c

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$ javac BitStuffing.java

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$ java BitStuffing
Bit Stuffing Mechanism
-----
Input Data : 01111101111110
Flag : 01111110
Stuffed Data : 0111110011111010111101001111110
Rule : Insert 0 after five consecutive 1s

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-32 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-33"

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-33 (main)
$ java -cp . FileClient
Connected to File Server.
File received successfully.
Saved as: received_sample.txt
Total bytes received: 40

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-33 (main)
$ cat received_sample.txt
This is a TCP File Transfer experiment.

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-33 (main)
$
```

Experiment – 34: CRC Error Detection / Cyclic Redundancy Check (CRC) Implementation

```
MINGW64/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-34

./Experiment-29: EchoClient.class EchoServer.class EchoServer.java
./Experiment-30: ChatClient.class ChatServer.class ChatServer.java
./Experiment-31: ARPImplementation.class ARPImplementation.java sample.txt
./Experiment-32: BitStuffing.class BitStuffing.java
./Experiment-33: FileClient.class FileServer.class received_sample.txt
FileClient.java FileServer.java sample.txt
./Experiment-34: CRCImplementation.class CRCImplementation.java
./Experiment-35: SlidingWindowClient.class SlidingWindowServer.class
SlidingWindowClient.java SlidingWindowServer.java

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-34"
$ java CRCImplementation
=====
CRC Error Detection Simulation
=====
Generator Polynomial : 0xED88320
Polynomial Bit Length: 32-bit (CRC-32)
=====

[1] Original Data Verification:
Original Data : "Computer Networks Lab Experiment 34"
Calculated CRC : 0x22E6D9A2
Expected CRC : 0x22E6D9A2
CRC Comparison : MATCH (Calculated == Expected)
CRC Verification : VALID

[2] Corrupted Data Verification:
Corrupted Data : "Computer Networks Lab Experiment 34"
Recalculated CRC : 0xB76A5F68
CRC Verification : ERROR DETECTED

=====
CRC simulation completed successfully.
=====

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-34 (main)
$
```

Experiment – 35: Implementation of Sliding Window Protocol

```
MINGW64/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C... 35"
jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-35 (main)
$ java SlidingWindowServer

=====
Sliding Window Server
=====
Sliding Window Server started...
Waiting for client...
Client connected: 127.0.0.1
-----
Received packet 0 with data: "FrameData-0"
Sending ACK for packet 0
Receiver window moved. Expecting window: [1, 2, 3, 4]
-----
Received packet 1 with data: "FrameData-1"
Sending ACK for packet 1
Receiver window moved. Expecting window: [2, 3, 4, 5]
-----
Received packet 2 with data: "FrameData-2"
Sending ACK for packet 2
Receiver window moved. Expecting window: [3, 4, 5, 6]
-----
Received packet 3 with data: "FrameData-3"
Sending ACK for packet 3
Receiver window moved. Expecting window: [4, 5, 6, 7]
-----
Received packet 4 with data: "FrameData-4"
Sending ACK for packet 4
Receiver window moved. Expecting window: [5, 6, 7]
-----
Received packet 5 with data: "FrameData-5"
Sending ACK for packet 5
Receiver window moved. Expecting window: [6, 7]
-----
Received packet 6 with data: "FrameData-6"
Sending ACK for packet 6
Receiver window moved. Expecting window: [7]
-----
Received packet 7 with data: "FrameData-7"
Sending ACK for packet 7
Receiver window moved. All expected packets received.
-----
All packets received successfully.
Sliding Window Server completed successfully.
=====

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-35 (main)
$

MINGW64/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C...
jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-35 (main)
$ cd "/e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-35"
$ java SlidingWindowClient

=====
Sliding Window Client
=====
Connected to Sliding Window Server.
Window Size: 4, Total Packets: 8
-----
Sending packet 0
Sending packet 1
Sending packet 2
Sending packet 3
ACK received for packet 0
Window slid
Current window: [1, 2, 3, 4]
Sending packet 4
ACK received for packet 1
Window slid
Current window: [2, 3, 4, 5]
Sending packet 5
ACK received for packet 2
Window slid
Current window: [3, 4, 5, 6]
Sending packet 6
ACK received for packet 3
Window slid
Current window: [4, 5, 6, 7]
Sending packet 7
ACK received for packet 4
Window slid
Current window: [5, 6, 7]
ACK received for packet 5
Window slid
Current window: [6, 7]
ACK received for packet 6
Window slid
Current window: [7]
ACK received for packet 7
Window slid
-----
All packets transmitted and acknowledged successfully.
=====

jaswa@DESKTOP-NM3SDRI MINGW64 /e/Development/College/Computer-Networks/CN-Lab-Experiments/Java-C/Experiment-35 (main)
$
```